

A Systematic Literature Review of Robust Federated Learning: Issues, Solutions, and Future Research Directions

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Federated Learning (FL) has emerged as a promising paradigm for training machine learning models across distributed devices while preserving their data privacy. However, the robustness of FL models against adversarial data and model attacks, noisy updates, and label-flipped data issues remain a critical concern. In this article, we present a systematic literature review using the PRISMA framework to comprehensively analyze existing research on robust FL. Through a rigorous selection process using six key databases (ACM Digital Library, IEEE Xplore, ScienceDirect, Springer, Web of Science, and Scopus), we identify and categorize 244 studies into eight themes of ensuring robustness in FL: objective regularization, optimizer modification, differential privacy employment, additional dataset requirement and decentralization orchestration, manifold, client selection, new aggregation algorithms, and aggregation hyperparameter tuning. We synthesize the findings from these themes, highlighting the various approaches and their potential gaps proposed to enhance the robustness of FL models. Furthermore, we discuss future research directions, focusing on the potential of hybrid approaches, ensemble techniques, and adaptive mechanisms for addressing the challenges associated with robust FL. This review not only provides a comprehensive overview of the state-of-the-art in robust FL but also serves as a roadmap for researchers and practitioners seeking to advance the field and develop more robust and resilient FL systems.

CCS Concepts: • **Security and privacy** → **Software and application security**; **Systems security**; **Social aspects of security and privacy**;

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1 Introduction

With the advent of the intelligent era, an increasing number of diverse industries, including smart health [121], smart transportation [197], smart manufacturing [98], and smart education [177], have been capitalizing on the potent advantages of **Artificial Intelligence (AI)** to facilitate industry upgrading. Concurrently, as the digital revolution advances, human society has entered the realm of **Industrial Revolution 4.0 (IR 4.0)**. This era is characterized by augmented interconnections between individuals and the physical world through a proliferating human-machine interface interaction [263]. At the crux of propelling high-quality development within IR 4.0, the progress and breakthroughs in intelligent technologies assume a pivotal role in augmenting the overall intelligence quotient of human society [3, 12]. Consequently, **Machine Learning (ML)**, as the primary technique of AI, has garnered significant attention from both academia and industry, fostering an unprecedented flourishing of related learning technologies [9, 194].

ML aims to enable computer systems to automatically learn and improve through experiential data, blending computer science and statistical principles. Over the past decades, advancements in ML have led to four main paradigms: unsupervised, semi-supervised, supervised, and reinforcement learning [86]. Early ML methods primarily relied on unsupervised techniques, like clustering and anomaly detection, and supervised techniques, such as linear/logistic regression, decision trees, and random forests, for tasks such as house price prediction and digit recognition [21]. However, these methods often required manual feature engineering and task-specific objective functions, complicating development [35]. To overcome these challenges, **deep learning (DL)** emerged as a powerful supervised learning framework with enhanced learning capacity and simplified model design [86]. DL utilizes layered network architectures to learn abstract representations, particularly useful for complex data [37]. With advancements in computing hardware, such as GPUs, DL has achieved significant success in tasks such as face and speech recognition, autonomous driving, and virtual reality [164, 215]. Despite its strengths, DL's reliance on large labeled datasets remains a key limitation, exemplified by the 45 terabytes of data used to train GPT-3 [250], making data scarcity a critical obstacle for further progress.

The potential remedy to address the scarcity of data for DL is found in the rapid proliferation of interconnected **Internet of Things (IoT)** devices, smartphones, and a diverse array of websites. Each individual end device has the capacity to function as a distinct data source, contributing a substantial volume of data on a daily basis, thereby leading to the emergence of “big data” in a distributed form [217]. This phenomenon has attracted the attention of ML researchers, prompting the development of distributed learning methodologies. Within the traditional distributed learning framework, one or multiple computing nodes, referred to as servers, aggregate the data generated by devices within the distributed network and subsequently undertake centralized training and inference of ML algorithms [28]. The primary advantage of this distributed learning paradigm lies in its potential to access a significant amount of data distributed across numerous end devices. However, this approach is not without its limitations, encompassing the burden of heavy storage and computational workloads imposed on servers, elevated communication costs entailed in transferring data between the server and devices, and, most notably, the inherent risk of data leakage [182].

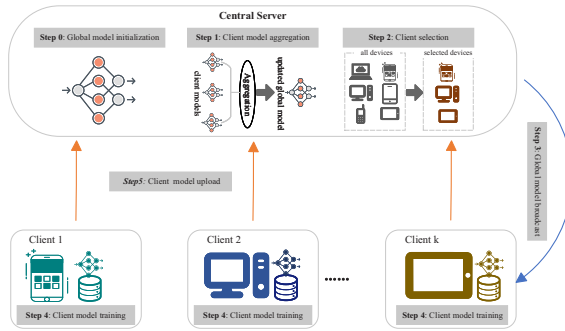


Fig. 1. Overview of the standard FL workflow predominantly comprising five key steps. Step 0 serves as the initial phase, occurring at the inception of FL. The subsequent processes from Step 1 to Step 5 entail iterative repetitions until the global model achieves a satisfactory convergence state.

The escalating frequency of data breaches in recent years has sparked significant concerns among users and government regulatory bodies [31]. Consequently, a series of laws and regulations, such as the European Union’s **General Data Protection Regulation (GDPR)** [75] and the United States’ **California Consumer Privacy Act (CCPA)** [13], have been enacted to safeguard users’ privacy and protect sensitive data. However, while these legislative measures aim to enhance data security, they have inadvertently created challenges in the collection of data from smart devices or particular institutions. This situation has resulted in the emergence of a so-called “data island” phenomenon, where private data stored in decentralized end devices becomes difficult to access for AI researchers, thereby impeding intelligent development across various industries [136]. As such, there arises an urgent necessity for the implementation of innovative distributed and decentralized learning technologies to effectively address the prevalent “data hungry” predicament faced within the AI research and application community [1].

To reconcile the contradiction between “data island” and “data hungry” challenges, a natural approach is to leverage the computational capabilities of edge devices for conducting on-device model training locally. This strategy obviates the need to transmit privacy-sensitive data between local devices and a central server, consequently reducing the risk of privacy infringement. Therefore, to enable access to data generated by private edge devices within the IoT, without compromising user privacy, a novel communication-efficient and privacy-preserving decentralized ML framework cited as **Federated Learning (FL)** was introduced in Reference [161]. In FL, a multitude of devices within the federated network collectively participate in the learning process of a globally shared deep model, coordinated by a trusted server. The objective is to ensure that the learned global model benefits each client device. The fundamental workflow of FL, as depicted in Figure 1, unfolds as follows: Initially, the server initializes a global model and broadcasts it to randomly selected available client devices. Upon receiving the global model, each client utilizes it to initialize its local model and performs iterative learning updates on its own private data. Subsequently, the updated local models are transmitted back to the server. Finally, the server employs an aggregation method to fuse all client models into a new global model. These iterative joint learning steps are repeated across multiple communication rounds until the final global model attains satisfactory performance.

FL has emerged as a promising research direction in the field of distributed DL, distinguishing itself from conventional distributed learning through two key features. First, FL prioritizes users’ privacy protection by restricting access to client raw data, in contrast to conventional distributed learning, which primarily focuses on accelerating model training. Second, FL shifts the entirety of

model training to individual end devices, whereas distributed learning heavily relies on the computational and storage resources of a central server. Leveraging the distribution of model training across diverse edge devices offers several advantages, including efficient reduction of communication costs, enhanced data privacy and security, optimal utilization of network device resources, and the facilitation of inter/cross-domain knowledge sharing. Owing to these attributes, FL has garnered favor among privacy-sensitive industries such as healthcare, social media, and transportation, where data is dispersed across geographically isolated devices [73, 76, 127, 195, 258]. Its potential applications encompass a wide range of domains, such as human activity recognition [10, 29], disease prediction [109, 180], sentiment analysis [191], and so on. Beyond applications within the same domain, researchers have explored the utilization of a novel integrated technique known as federated transfer learning, aimed at facilitating cross-domain knowledge sharing and collaboration. This becomes particularly relevant when there is minimal overlap in either users or user features between datasets originating from disparate domains [92, 143].

While FL presents a remarkable distributed learning paradigm that upholds data privacy, it encounters preliminary issues arising from its distinctive operational approach. First, FL's primary objective is to train a global model that caters to each federated participant. However, unlike conventional distributed learning, which enjoys comprehensive access to data distribution across clients, FL faces the limitation of incomplete perception of such global data distribution. This characteristic introduces heightened optimization issues for FL algorithms. Of particular significance is the statistical heterogeneity of federated data distribution, stemming from the heterogeneity of participating devices with varying storage and computation capabilities, as well as diverse user operation habits. Such statistical heterogeneity undermines the preliminary assumption of independently and **identically distributed (IID)** data, presenting a hurdle for traditional ML optimization. Second, although raw data transfer is unnecessary in the federated network, multiple communication rounds are still indispensable to exchange model gradients between clients and the server. These rounds ensure the synchronization of global model gradient updates and the achievement of global model convergence. Consequently, elevated communication costs are a significant obstacle to the practical application of FL. As the aforementioned issues are the preliminary challenges in the FL scheme, numerous works have been done on their alleviation from the emergence of FL (see the following surveys: [87, 93, 133, 158, 171, 185, 192, 245, 275, 276, 282, 316, 317] for detailed insights).

To address these inherent challenges in FL, researchers have turned to the development of robust FL [56] that aims to strengthen the FL framework by enhancing its resilience against adversarial threats and unreliable participants. Particularly, robust FL seeks to mitigate the risks posed by malicious clients, noisy updates, and inconsistent data by employing advanced aggregation methods, anomaly detection, and robust optimization techniques [46]. By incorporating robustness into the FL paradigm, researchers aim to create models that can maintain high accuracy and reliability, even in hostile or unpredictable environments. This addition of robustness is critical for real-world applications of FL, where adversarial conditions are not only possible but increasingly likely [36].

Presently, the robustness challenge in FL has become a burning issue that arises due to the decentralized nature of the training process, where multiple client devices collaboratively contribute to model training keeping their data locally and the FL server lacks knowledge of the client's training process, potential malicious attacks on client data, client models, and its training process. This collaborative setup introduces several vulnerabilities and threats that can undermine the integrity and performance of the globally trained model. Specifically, the challenge encompasses various factors as follows: (i) Adversarial data and model attacks: Clients can be compromised or malicious, introducing adversarial data that is intentionally crafted to deceive the model. Similarly, malicious clients can contribute harmful model updates to compromise the global model's performance or

inject malicious code. (ii) Noisy updates: The training data from clients might contain noise due to various reasons, such as measurement errors or data transmission issues. (iii) Label-flipped data: Mislabeling of data by clients can lead to inconsistencies and confusion during the aggregation process. This is particularly challenging when different clients have varying label distributions. This noise can lead to unreliable model updates, affecting the convergence and overall model quality. We contend that fortifying FL against malicious attacks constitutes a compelling research trajectory to establish robustness for real-world applications. Consequently, this article shall delve into existing technical advancements, thereby facilitating the delineation of a comprehensive panorama of robust FL. This, in turn, shall steer more researchers toward exploring the tenets of FL robustness.

As a seminal work in this area, **Federated Averaging (FedAvg)** [161] introduced the concept of increasing local training and adopting weighted averaging for client models within the **stochastic gradient descent (SGD)**-based federated training framework. However, FedAvg demonstrates suboptimal performance when confronted with non-IID data, as subsequent FL studies have experimentally and theoretically demonstrated its unstable convergence or even divergence characteristics under highly skewed non-IID scenarios [123, 307]. Moreover, FedAvg-based methods are susceptible to malicious attacks, particularly when adversaries impersonate benign clients to participate in joint training. These malicious clients can launch various attacks, such as contaminating labels or features of raw data and tampering with client gradient updates, thereby compromising the utility of the global model [45, 107, 209, 227, 285]. It is noteworthy that conventional attack detection methods designed for centralized ML are inapplicable in FL settings due to the lack of access to client data by the FL server. Consequently, the quest for achieving robust FL has emerged as a prominent research focus within the FL community, serving as the primary concern of our study. Addressing these challenges requires innovative approaches that go beyond conventional ML techniques. Researchers are exploring a range of strategies, including modifying optimization algorithms, integrating privacy-preserving mechanisms, developing robust aggregation methods, and enhancing client selection strategies. These approaches collectively aim to enhance the robustness of FL models against adversarial attacks, noisy data, and other vulnerabilities, while maintaining the privacy and efficiency benefits of the decentralized learning paradigm.

To bolster the robustness of FL, several diverse strategies have been explored and attempted to address this challenge. A considerable portion of studies [67, 117, 131, 259, 274, 315] have concentrated on devising secure and robust model aggregation methods to effectively counter malicious attacks and accommodate heterogeneous data distribution. The commonly adopted element-wise weighted averaging approach has shown limitations in accurately distinguishing client contributions and credibility, leading to inadequate fairness and security in the fusion of client models. As a significant technical avenue, another approach involves modifying the client-side training scheme to align the local training objective with the global objective. This alignment aims to rectify issues related to client gradient drift or abnormal gradient updates [23, 40, 44, 66, 96, 323]. Additionally, few research works adopt a dual-pronged approach to further fortify the robustness of the FL optimization process [11, 110, 118, 184, 246, 313].

To the best of our knowledge, comprehensive surveys addressing the topic of robust FL are scarce. In comparison to existing surveys and reviews on robust FL, our SLR stands out in several ways. First, our SLR uses the **Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)** framework to meticulously select and categorize 244 studies into eight distinct themes. This exhaustive coverage ensures that we consider a wide range of approaches and research on robust FL. Second, we provide an in-depth analysis of each theme, discussing the various methods and techniques proposed to enhance the robustness of FL models and identifying their potential gaps. This detailed examination allows readers to gain a comprehensive understanding of the state-of-the-art in this field. Third, unlike many existing surveys, we go beyond

Table 1. Comparison between Our SLR and Prior Survey Efforts Reveals Its Comprehensiveness, Up-to-date Nature, and Incorporation of Diverse Viewpoints

Study	Year	Type of Survey	No. of Search Databases	Bibliometric Analysis	Thematic Analysis	No. of Studies	No. of Issues to Hamper Robustness	No. of Strategies to Attain Robustness
[213]	2022	Unstructured Review	1	No	No	8	1	2
[113]	2020	Unstructured Review	1	No	Yes	40	3	3
[289]	2022	Unstructured Survey	1	No	Yes	24	4	3
[153]	2020	Unstructured Survey	1	No	Yes	18	6	0
[152]	2022	Unstructured Survey	1	No	No	22	2	2
[186]	2021	SLR (not on robust FL)	-	-	-	-	-	-
[324]	2021	Unstructured Survey (not on robust FL)	-	-	-	-	-	-
[205]	2024	Unstructured Survey	1	No	Yes	33	3	6
[294]	2024	Unstructured Survey	1	No	Yes	31	3	3
[252]	2024	Unstructured Survey	1	No	Yes	39	2	4
[283]	2024	Unstructured Survey	1	No	Yes	46	3	2
[6]	2023	Unstructured Survey	1	No	No	14	2	2
[94]	2023	Unstructured Survey	1	No	No	50	1	-
Ours	2025	SLR	6	Yes	Yes	244	8	8

summarizing existing work. We discuss future research directions, highlighting the potential of hybrid approaches, ensemble techniques, and adaptive mechanisms. This forward-looking perspective is valuable for researchers and practitioners seeking to advance the field. Fourth, we include a bibliometric analysis as an additional contribution, shedding light on the trends and research landscape in robust FL. Overall, our SLR goes beyond existing surveys by offering a comprehensive, up-to-date, and multi-perspective overview of the robustness challenges and solutions in FL, as condensed in Table 1. To this end, we provide the list of contributions to our work as follows:

- **Comprehensive Literature Review:** While many survey papers provide a review of the literature, our work distinguishes itself by adhering to the rigorous PRISMA framework, which ensures systematic selection, categorization, and analysis of 246 studies, providing exhaustive coverage (2016.01.01 and 2024.08.26) across eight distinct themes related to robust FL. This structured approach contrasts with more generalized reviews, which often lack such comprehensive thematic analysis.
- **Bibliometric Analysis:** Our inclusion of bibliometric analysis goes beyond simply categorizing studies. It provides insights into trends such as the year-wise publication trends, geographical distribution of research, and key contributors in the field. This analysis offers unique value by identifying emerging research hubs and collaboration patterns that are typically overlooked in conventional survey papers.
- **Categorization of Research Themes:** Unlike many surveys that merely summarize research efforts, we categorize the selected studies into eight themes of ensuring robustness in FL, namely, Objective Regularization, Optimizer Modification, Differential Privacy Employment, Additional Dataset Requirement and Decentralization Orchestration, Client Selection, New Aggregation Algorithms, Aggregation Hyperparameter Tuning, and Hyperparameter Tuning during Model Aggregation. This thematic organization provides a novel way for researchers to navigate the diverse solutions proposed in the field and easily identify methods relevant to their specific use cases.
- **Synthesized Thematic Findings:** The synthesis of findings within each theme, accompanied by potential research gaps, sets our work apart by offering a holistic view of the strengths and limitations of existing FL techniques. This approach goes beyond a summary of prior works, providing a roadmap for how future research can address the identified gaps.
- **Future Research Directions:** While it is common for survey papers to propose future research directions, our review identifies concrete and underexplored areas, such as hybrid approaches, ensemble techniques, and adaptive mechanisms. This forward-looking analysis

is grounded in a critical assessment of the literature, offering a uniquely actionable set of future challenges and opportunities for robust FL.

The rest of the article is organized as follows: In Section 3, we describe the methodology used for conducting the SLR. We introduce the PRISMA framework, which guides the review process, including study selection, data extraction, and categorization. Section 4 analyzes the bibliometric descriptive results of the selected studies in the SLR. We present the results of the SLR by categorizing the selected studies into eight thematic areas in Section 5. Building upon the synthesized thematic findings, Section 6 outlines future research directions to enhance the robustness of FL models. The article concludes by summarizing the contributions of the SLR in Section 7.

2 Preliminaries of Robust FL

This section introduces the fundamental concepts of robust FL, covers the key challenges, adversarial attacks and noisy updates, and outlines evaluation metrics to assess performance. Additionally, real-world case studies are presented to highlight the practical applications of robust FL.

2.1 Introduction to Robust FL

Robust FL refers to an enhancement of FL that focuses on ensuring the security and stability of models trained across decentralized clients. Robust FL aims to address the inherent vulnerabilities in FL, particularly in environments where data privacy is critical but malicious or unreliable behavior by some clients can undermine the model's integrity [36, 46, 56]. By improving the system's resilience against adversarial attacks, noisy updates, and corrupted labels, robust FL ensures that the global model can maintain high performance and security even when faced with unreliable participants. Robust FL encounters several specific challenges that threaten its robustness. **Adversarial Data and Model Attacks:** Malicious clients may deliberately manipulate their local data or model updates to degrade the performance of the global model. This includes injecting poisoned data or manipulating gradients to skew the aggregation process, making it harder for the model to converge to a reliable state [46, 295]. **Noisy Updates:** Clients may unintentionally send noisy or unreliable updates due to factors such as limited computational resources, unstable network conditions, or inaccurate local training. These noisy updates can lead to suboptimal model performance if not properly handled during aggregation [27, 46]. **Label-flipped Data:** Some clients may possess or deliberately introduce label-flipped data, where correct labels are intentionally or mistakenly altered. This can significantly impact the model's ability to generalize and result in poor decision-making, particularly in critical applications such as healthcare or finance [80, 84]. By addressing these challenges, robust FL aims to create a more secure, reliable FL system that is robust to both intentional adversarial activities and accidental client-side errors.

2.2 Evaluation Metrics for Robust FL

In robust FL, several quantifiable metrics are used to evaluate the system's performance and security. Testing accuracy is measured by the federated global model's ability to correctly predict outcomes on a validation set. However, robustness is typically assessed by testing the federated global model under adversarial conditions, such as injecting poisoned updates or noisy data, and observing the degradation in performance [287]. A key metric here is robust accuracy, which compares the global model's performance when all clients are benign versus when some clients are adversarial. The difference in performance, known as robustness training loss, helps quantify the system's resilience to attacks [74]. Privacy leakage is measured using techniques like differential privacy, where the privacy budget, denoted by ϵ , defines the upper limit on how much an adversary can infer about any client's data [189]. Convergence rates are tracked by counting the number of

communication rounds or iterations needed for the federated model to stabilize or achieve a target accuracy, both under normal and adversarial conditions [130]. These metrics together offer a clear understanding of the robustness and security of robust FL systems [78, 294].

2.3 Case Studies in Robust FL

The real-world applications of robust FL demonstrate its critical role in various sectors where privacy, security, and robustness are essential. Robust FL enables organizations to train ML models collaboratively while ensuring that individual data remains private and the model is resilient against adversarial attacks, unreliable updates, and data corruption. Below, we explore how robust FL is being utilized in practical settings.

In the healthcare industry, privacy and security of sensitive patient data are of utmost importance. Robust FL allows multiple healthcare institutions, such as hospitals and research centers, to collaboratively train predictive models on patient data without exposing the data itself. This is particularly beneficial for tasks like disease diagnosis, where models can be trained using a wider variety of data from different hospitals, improving accuracy and generalization. At the same time, robust FL ensures that adversarial clients—whether intentionally malicious or simply faulty—cannot compromise the model’s integrity. This balance of data privacy and robustness makes robust FL an ideal solution for sensitive applications in healthcare [97, 172]. In autonomous vehicle systems, large-scale ML models are crucial for decision-making in complex and dynamic environments. Robust FL provides a way for manufacturers to jointly develop these models by sharing model updates rather than proprietary data, ensuring intellectual property protection. However, the potential for adversarial attacks, such as model poisoning, poses a serious threat in this domain. A compromised global model could lead to catastrophic failures in autonomous vehicle behavior. Robust FL mitigates this risk by ensuring the system’s resilience to such attacks, allowing autonomous vehicle models to operate reliably even when facing adversarial clients [247, 310].

The implications of robust FL extend beyond healthcare and autonomous vehicles. As decentralized learning systems become more widespread in industries such as finance [170], **Internet of Things (IoT) networks** [171], and smart cities [178], the ability to secure these systems against adversarial actions while preserving privacy becomes crucial. Robust FL offers a path forward by ensuring that models can be trained securely and efficiently in environments where data security, privacy, and robustness are non-negotiable. This makes robust FL a foundational technology for future applications that rely on secure and scalable distributed learning systems. To this end, robust FL presents a powerful solution for addressing the security and robustness challenges inherent in decentralized learning environments. By improving resilience against adversarial attacks, noisy updates, and faulty clients, robust FL ensures reliable performance and data privacy across various sectors. From healthcare and autonomous vehicles to finance and IoT networks, robust FL is poised to become a foundational technology for secure, decentralized learning systems that prioritize both privacy and robustness. Future developments in this field will continue to expand its applicability and strengthen the security of collaborative learning environments.

3 Methodology

The methodology for our SLR involves a structured and systematic approach to identify, select, and analyze relevant research studies on the topic of robust FL. The PRISMA framework guides the methodology, ensuring transparency and rigor in the review process.

3.1 Search String Definition

To expand the search results, we utilize “Robust Federated Learning” as the key term and include synonyms and acronyms as supplemental terms. For each primary source, we create search strings

Table 2. Search Key and Supplementary Terms

Key Term	Supplementary Terms
Robust Federated Learning	robust federated aggregation, byzantine-robust federated learning, byzantine-robust aggregation, byzantine-resilient federated learning, byzantine-resilient aggregation, outlier-robust federated learning, outlier-robust aggregation, outlier-resilient federated learning, outlier-resilient aggregation, attack-adaptive aggregation, attack-adaptive federated learning, backdoor federated learning.

that check the title, abstract, and keywords. After finishing the first draft of search strings, we check the results of each search string against each database to determine their effectiveness. Table 2 displays the finalized search terms.

3.2 Research Question

The primary aim of this SLR is to comprehensively investigate and synthesize the existing body of research on Robust FL. Specifically, we seek to address the following research questions: (i) What are the current state-of-the-art mechanisms to hamper the robustness of FL? (ii) What are the present state-of-the-art approaches in achieving robustness in FL? (iii) What are the potential research gaps of the state-of-the-art methods in attaining robustness in FL? and (iv) What would be the open problems and promising future research directions toward obtaining a robust FL framework?

3.3 Study Design

This SLR adheres to the PRISMA framework, a well-established guideline for conducting systematic reviews. The PRISMA framework ensures a transparent, comprehensive, and reproducible process, enabling us to identify, select, and synthesize relevant studies effectively.

3.4 Study Source

To identify relevant studies for this review, a systematic search has been conducted across various academic databases and publication repositories. Key databases included are (i) ACM Digital Library, (ii) IEEE Xplore, (iii) ScienceDirect, (iv) Springer, (v) **Web of Science (WoS)**, and (vi) Scopus. The search timeframe is set in between 2016.01.01 and 2024.08.26. We screen and select the papers from the initial search according to the preset inclusion and exclusion criteria elaborated in Section 3.6.

3.5 Search Strategy

A comprehensive search strategy has been formulated using a combination of keywords and controlled vocabulary relevant to Robust FL. We tailor the search strings to the specific syntax of each database. Boolean operators (AND and/or OR) and truncation symbols are used as necessary to ensure comprehensive coverage of relevant literature.

3.6 Eligibility Criteria

To ensure the inclusion of high-quality and pertinent studies, the following eligibility criteria have been applied during the study screening and selection process:

- Inclusion criteria:
 - Studies published in peer-reviewed journals, conference proceedings, and reputable sources.
 - Studies that focus on Robust FL, addressing methods, techniques, algorithms, and applications.

Table 3. Search Result Quantity and Selected Studies of Different Databases

Database	Result quantity	Selected papers	Database	Result quantity	Selected papers
ACM	110	37	Springer	236	27
IEEE Xplore	205	159	WoS	132	104
ScienceDirect	90	45	Scopus	221	153
Total unique selected studies: 244					

- Studies that present empirical research, experiments, simulations, or case studies related to attaining robustness in FL.
- Studies published in the English language.
- Exclusion criteria:
 - Studies that focus only on hampering robustness in FL without proposing any potential solutions to alleviate it.
 - Survey, review, and SLR studies, as they do not contribute to the development of attainment of robustness in FL.
 - Studies that are not in English.
 - Academic dissertations, tutorials, editorials, and magazines.

3.7 Study Selection

We perform the selection process in two phases: title/abstract screening and full-text screening. Two independent researchers screened the titles and abstracts of the retrieved studies against the eligibility criteria to identify potentially relevant articles. Discrepancies are resolved through discussion or consultation with a third researcher. Subsequently, the full texts of the selected articles are assessed for final inclusion. As a result, we found a total of 244 unique studies, while we provide the amounts of the initial and selected studies for each primary source in Table 3. Notice that the sum of selected studies from these databases exceeds 244, as some of these selected studies have appeared in multiple databases. It is worth mentioning that a significant portion of these papers were initially found in Scopus and subsequently identified in other databases as well.

3.8 Data Extraction and Synthesis

Data extraction is performed by two independent researchers using a standardized data extraction form. The form captures essential information from each included study, such as authors, publication year, type of robustness considered, modifying FL component for attaining the robustness, robustness technique and approach, evaluation type, application domain, and dataset used. In addition, we find out the potential gap(s) of every included study to be considered by the research community. All data are recorded in the Microsoft Excel sheet for analysis and synthesis processes. The extracted data are synthesized and summarized qualitatively to address the research questions.

3.9 Reporting

The results are reported following the PRISMA guidelines. The findings are organized based on eight (08) themes and research trends. We include a flow diagram in Figure 2 depicting the study selection process, a summary of key findings, and an in-depth discussion of the research trends, challenges, and opportunities in Robust FL considering the aforementioned eight themes.

3.10 Discussion and Future Directions

The findings are discussed in the context of the existing literature. Emerging trends and gaps in the field are identified, leading to the formulation of future research directions to address the challenges of robust FL.

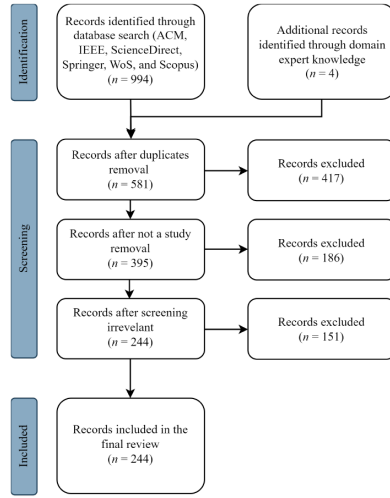


Fig. 2. RRISMA flow diagram of the study selection process.

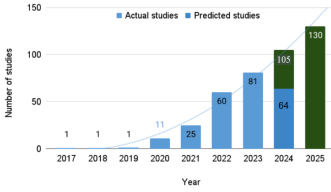


Fig. 3. Number of yearly publications and prediction of the future years.

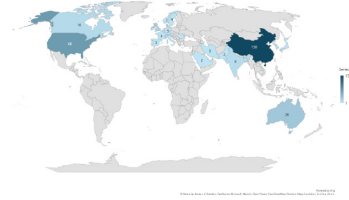


Fig. 4. Geographical scope of the studies.

4 Descriptive Result Analysis

4.1 Publication Trend

We utilize graphs and charts in our bibliometric analysis to illustrate the findings of scientific endeavors documented in research databases. Figure 3 illustrates the trajectory of scholarly publications concerning the attainment of robustness in FL spanning the period from 2017 to August 2024, with a prediction for the remaining of 2024 and whole 2025, using the Linear regression [59]. This graphical representation indicates an upward trajectory in the number of publications addressing this subject. Before 2020, there was a notable absence of substantial research focusing on the robustness associated with FL. This can be attributed partly to the fact that robustness challenges in FL only emerge after the initial stages of the adoption and execution of FL. The primary surge in research regarding these challenges occurred between 2020 and 2022. By 2022, the volume of publications concerning the robustness associated with FL had increased significantly. We have even more papers from 2023 and a substantial number of papers from 2024 (up until August). This surge can be attributed to the nascent stage of successful usage and execution of FL globally.

4.2 Coverage of Researchers

Figure 4 portrays the distribution of publications across different countries concerning the attainment of robustness in FL, while Table 4 provides the country names and their actual number of publications. A significant portion of research efforts in this domain has been concentrated in China.

Table 4. Country-wise Authors Count of the Selected Studies

Label	Country	Studies	Label	Country	Studies	Label	Country	Studies
1	China	138	11	Spain	3	21	Norway	2
2	USA	66	12	Qatar	3	22	Netherlands	4
3	Australia	26	13	Iran	3	23	KSA	2
4	UK	19	14	India	9	24	Italy	4
5	Singapore	8	15	Germany	5	25	Japan	3
6	Hong Kong	8	16	Austria	3	26	Romania	1
7	South Korea	9	17	Pakistan	2	27	Ireland	1
8	Canada	10	18	UAE	4	28	Bangladesh	1
9	Sweden	4	19	France	4			
10	Switzerland	3	20	Luxembourg	1			

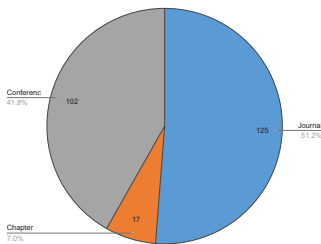


Fig. 5. Types of the studies from 2017 to 2024.

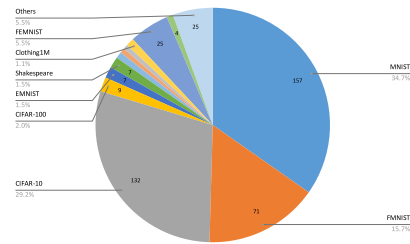


Fig. 6. Commonly used datasets from 2017 to 2024.

Up until August 2024, Chinese researchers had authored 138 publications in this field, while the United States has made relatively comparable contributions to the literature.

4.3 Type and Application of the Studies

Figure 5 illustrates the predominant types of publications, namely, conference articles (42%), journal articles (51%), and book chapters (7%), during the period from 2016 to August 2024. This data suggests that researchers and scholars exhibit a stronger inclination toward disseminating their work through participation in conferences and journal articles. These platforms offer the opportunity to receive targeted recommendations, insights, and feedback from reviewers and fellow participants. However, Figure 6 provides an overview of publications categorized by different application areas, addressing challenges related to robustness in FL from 2016 to 2024. The data from Figure 6 underscores the widespread concern among researchers from different application domains regarding robustness issues with FL. **Computer Vision (CV)**, language modeling, and face recognition received the most attention among the application areas. Conversely, application areas such as cybersecurity, IoT, health, and bioinformatics exhibit limited research output. Consequently, this discovery highlights the need for increased research focus within these application domains.

4.4 Collaborative and Non-collaborative Nature of the Studies

Figure 7 depicts the sources (organizations) of the research conducted. It can be seen that 88.9% of the research was independently undertaken by universities, with a smaller fraction originating from industrial entities and academia-industry collaborations. However, Figure 8 visualizes the network of co-authorship connections between countries in the field of robustness in FL. In this

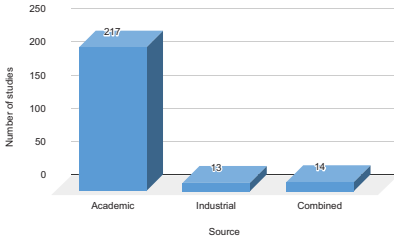


Fig. 7. Source of the studies from 2017 to 2024.

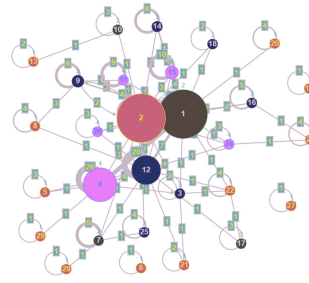


Fig. 8. Co-authorship network of countries.

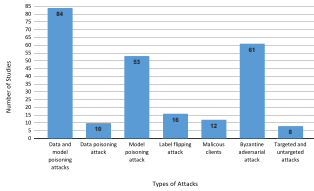


Fig. 9. Distribution of the studies based on different attacks.

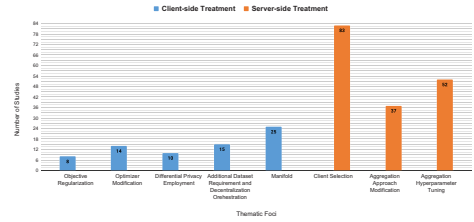


Fig. 10. Distribution of the studies based on thematic foci.

network, each node represents a country, and the node's size is based on the number of publications from that country. The thickness of the connections between nodes indicates the strength of collaboration between those countries. The map depicted in Figure 8 illustrates the central collaborative roles of China, the United States, the United Kingdom, and Australia, as indicated by their larger nodes. Singapore, Hong Kong, South Korea, India, and Canada show moderate levels of collaboration with other countries. Conversely, Austria, Luxembourg, France, Kingdom of Saudi Arabia, Ireland, Norway, and Bangladesh emerge as less central countries in terms of co-authorship. Notably, the intensified connections between China and the United States suggest a robust collaborative partnership between these two nations.

4.5 Distribution of the Studies Based on Different Attacks

A total of 244 studies encompass seven distinct types of attacks. Data and model poisoning attacks account for 84 studies, Byzantine adversarial attacks account for 61 studies, and model poisoning attacks comprise 53 studies. These three categories collectively constitute a substantial portion of the studies. Additionally, 10 studies each are dedicated to data poisoning attacks, and 16 studies on label-flipping attacks. The class of malicious clients encompasses 12 studies, leaving the remaining 8 studies categorized under targeted and untargeted attacks, visually represented in Figure 9.

4.6 Distribution of the Studies Based on the Eight Thematic Foci

A total of 244 studies have been categorized across eight themes that we consider representative of ensuring robustness in FL. Out of these 244 studies, 72 pertain to client-side treatment, while the remaining 172 are related to server-side treatment. Within the client-side treatment category, 8 studies focus on objective regularization, 14 on optimizer modification, 10 on the implementation of differential privacy, 15 on additional dataset requirements and decentralization orchestration, and the remaining 25 are centered around manifold methods. On the server-side treatment, the majority of studies, totaling 83, revolve around aggregation hyperparameter tuning, followed by

37 studies concerning client selection, and 52 studies involving modifications to the aggregation approach. These findings are visually represented in Figure 10.

5 Thematic Result Analysis

The encompassing review of selected studies ($n = 244$) investigates diverse dimensions of techniques to attain robustness in FL. To systematically synthesize this extensive collection of findings, each study underwent meticulous evaluation to unveil recurring themes. Employing content analysis, we identify eight key emergent themes of ensuring robustness in FL; five belong to the client-side treatment and the rest (three) belong to the server-side treatment, for obtaining robustness in FL ensuring that the thematic foci would present an unbiased and clear narrative of the selected studies. In particular, the eight themes are (i) Objective Regularization, (ii) Optimizer Modification, (iii) Differential Privacy Employment, (iv) Additional Dataset Requirement and Decentralization Orchestration, (v) Manifold, (vi) Client Selection, (vii) Aggregation Approach Modification, and (viii) Aggregation Hyperparameter Tuning.

5.1 Client-side Treatment

5.1.1 Objective Regularization. Objective regularization methods in FL aim to enhance robustness by introducing additional terms or penalties into the optimization objective function. These approaches focus on improving the global model's performance and stability by encouraging better cooperation among participating clients while considering the potential presence of malicious or faulty data. In this SLR, we identified eight works focusing on objective regularization to ensure robustness in FL using different process.

L2 and Norm-based Regularization: L2 regularization is commonly used to improve model robustness by preventing overfitting. Probabilistic L2 norm checks are applied in RiseFL [322] to resist Byzantine attacks, although these checks can be computationally costly. FedProx [120] also utilizes L2 regularization, ensuring consistent model updates across clients to stabilize global model performance. **Robustness through Jacobian and Huber Regularization:** Jacobian-based regularization is employed in Flmjr [66] to maintain model stability by constraining perturbations in the model structure. Meanwhile, multi-dimensional Huber loss minimization [305] promotes Byzantine robustness, achieving promising results, though non-IID data scenarios require further testing. **Loss Function Approximation Techniques:** Loss function-based regularization can prevent models from becoming overfitted to local noise. Convex approximation and loss function smoothing [8] leverage a sampling-based convex approximation to optimize the model objective function, especially effective under non-IID client data distributions.

Label Noise Handling through Output Regularization: Regularization methods targeting noisy labels are crucial for managing variability in client data quality. In Reference [82], self-distillation is used to minimize output discrepancies between original and augmented data, reducing the model's tendency to memorize noise. Local weight adjustment on noisy datasets [27] enables edge devices to adjust local weights before aggregation, improving cooperation between clients and central servers. **Adaptive and Byzantine-resilient Loss Selection:** Adaptive approaches to regularization allow the model to respond dynamically to malicious data. Adaptive regularization with loss selection in Reference [101] combines Byzantine-resilient aggregation with loss-based adjustments, which aids intrusion detection but may face challenges with model stability over time. A comprehensive comparative analysis of these approaches is provided in Table 5, which highlights potential gaps in their effectiveness.

5.1.2 Optimizer Modification. Approaches that focus on modifying optimization methods, such as SGD, Adam, and others, play a crucial role in enhancing the robustness of FL. These methods

aim to address challenges associated with noisy updates and adversarial attacks, which can hinder convergence and degrade model performance. This theme in our SLR comprises 14 studies, summarized as follows:

Byzantine-resilient Gradient Descent: Several approaches enhance SGD to withstand Byzantine attacks. Byzantine-resilient SGD [200] simplifies convergence theory to improve resilience, while adaptive gradient descent [224] combines trimmed mean aggregation for robustness in text classification. Additionally, maximum transmit power adjustments [44] in SGD enhance update reliability, and gradient-similarity-based secure consensus [261] dynamically excludes Byzantine gradients. **Noise and Compression Reduction Techniques:** Noise reduction and compression improvements focus on maintaining gradient integrity during transmission. Gradient difference compression [315] reduces compression noise, boosting Byzantine resilience, while ternary quantization and error feedback [265] enhance the detection of malicious clients. PolyFlagSVM [165] applies encryption for noise resilience, particularly in medical imaging. **Robust Optimization with Minimax and Distributional Methods:** Robust optimizations include distributionally robust optimization [211] for noisy-label resilience in intrusion detection, albeit with added complexity. Minimax optimization [198] guards against distribution shifts, enhancing general robustness but requiring computational tradeoffs.

Resampling and Buffering Techniques: Resampling methods address variations in updates from Byzantine clients. SAGA [184] applies a resample-then-aggregate strategy to counter Byzantine variations. Buffered asynchronous gradient descent [269] utilizes gradient buffers and adaptive reassignment, which are effective against poisoning attacks but can delay convergence. **Adaptive Optimization and Client Selection:** Adaptive methods modify optimization strategies based on current client behavior. Adaptive FL [230] adjusts to model poisoning and Byzantine threats, though effectiveness can decline under high attack intensities. Wasserstein-distance-based client selection [204] strengthens adversarial robustness by dynamically selecting clients, despite resource demands. **Lightweight and Computation-optimized Methods:** Optimizations that reduce computation focus on minimizing load for non-faulty agents. For instance, non-faulty agent optimization [137] lowers computational demand by prioritizing essential tasks. Stochastic gradient methods [261] apply gradient selection strategies to minimize unnecessary calculations, supporting scalability. These approaches are compared comprehensively in Table 6 to improve FL robustness against noisy and adversarial attacks.

5.1.3 Differential Privacy Employment. Approaches that leverage differential privacy to enhance the robustness of FL focus on preserving individual privacy while aggregating model updates from multiple clients. These methods address data privacy and security concerns, ensuring that the aggregated global model remains resilient against privacy breaches and adversarial attacks. By introducing controlled noise to the model updates, differential privacy techniques offer a formal privacy guarantee while enabling effective collaborative learning. In our SLR, we identified 10 studies within this theme, which are discussed below.

Basic Differential Privacy Mechanisms for Privacy Preservation: Some approaches employ traditional DP to secure model updates. Joint DP in FL [290] limits adversarial influence by ensuring each update minimally impacts the global model, while **Local Differential Privacy (LDP)** [148] adds local noise before updates are aggregated, improving privacy protection. **Shuffle and Protocol-based Privacy Schemes:** Protocols using shuffling enhance privacy by obscuring individual contributions. In Reference [156], shuffle protocols and summation-based DP secure model parameters during exchange and global aggregation. SIREN+ framework [65] combines DP with proactive alarms, offering Byzantine resilience but with high communication overhead. **Gaussian Noise and Variance Reduction Techniques:** Gaussian noise is added to model updates in

schemes like VPPFL [79], defending against model poisoning with DP-backed noise injection. Sparsification and variance reduction [297] merge DP with sparsity to minimize communication loads, yet this approach may lead to high variance and processing demands in large models.

Game Theory and Adversarial Training Integrations: Integrating game theory and adversarial training into DP frameworks adds robustness against targeted attacks. Adversarial training with logit calibration [190] manages class imbalances, though it struggles with scalability. Game theory and DP integration [290] protects against model poisoning, balancing client influence with privacy, though practical challenges arise in implementation. **DP-enhanced Gradient Descent and Online Learning:** Combining DP with gradient descent techniques provides robust learning against Byzantine clients. Online Mirror Descent with DP [176] offers a solution for continuous learning under DP, but it faces scalability limitations. In Reference [220], weak DP in gradient descent tackles frequent, unconstrained attacks, securing updates during random sampling. To this end, we provide a comprehensive comparative unit analysis indicating the potential gaps in these approaches in Table 7.

5.1.4 Additional Dataset Requirement and Decentralization Orchestration. Approaches that involve additional dataset requirements and decentralized orchestration aim to enhance the robustness of FL by considering diverse data sources and improving coordination among participating clients. These methods address challenges related to noisy updates and adversarial attacks, which can hinder convergence and degrade model performance in decentralized settings. In our SLR, we identified 15 studies within this theme.

Blockchain and Smart Contract Integration: Blockchain technology provides a decentralized ledger that enhances data integrity, resilience, and security within FL. Studies such as Reference [246] present a blockchained decentralized FL framework for IoT to enhance privacy and defend against poisoning attacks and server failures. However, it is limited to MNIST testing without new aggregation methods, non-IID data considerations, or robust baseline comparisons. Additionally, blockchain-based FL for privacy and data integrity [46] uses blockchain to counter poisoning and server failures, while FedAnil [49] combines blockchain with clustering and encryption, despite facing non-IID data limitations. Blockchain-based consensus with smart contracts [48] incorporates differential privacy, although scalability remains a constraint. **Shadow and Reference Dataset Use for Model Evaluation:** Shadow datasets are used as external references to evaluate or filter out malicious models. In FedSuper [304], a **Local Model Filter (LMF)** applies shadow data for poisoning detection, while shadow datasets in filtering [243] employ accuracy metrics for model evaluation, though both approaches encounter scalability issues. Reference datasets [253] allow nodes to assess model performance, adding robustness against adversarial updates.

Cross-validation and Auditing Mechanisms: Cross-validation techniques support model evaluation in FL, often paired with decentralized or blockchain setups. Cross-validation against Byzantine attacks [111] uses blockchain validation but faces computational overhead. FL-Auditor [299] introduces third-party auditing to increase accountability, though scalability is a concern. **Decentralized Orchestration for Improved Resilience:** Decentralized orchestration approaches reduce reliance on a central server by distributing roles among clients. Ring-Allreduce with secret sharing [226] enables robust, decentralized aggregation, though communication demands are high. Byzantine resilience through role rotation [16] further enhances robustness, though it requires testing across different attack types. **Privacy-preserving Techniques in Decentralized FL:** Privacy-preserving methods in decentralized FL combine blockchain or other security techniques with privacy measures. Hyperledger Fabric for secure predictions [89] protects FL data, while Cre-DPSIGN with differential privacy [39] combines blockchain and DP but requires extensive tuning. Reverse aggregation [103] and gradient aggregation with anomaly detection [141]

add privacy layers but can suffer from overhead and complexity. We provide a comprehensive comparative analysis, outlining the potential gaps in these approaches in Table 8.

5.1.5 Manifold. Manifold approaches in FL address robustness by leveraging manifold properties to enhance resilience against adversarial attacks, noise, and data heterogeneity. These 25 studies ([22, 23, 40, 47, 54, 72, 81, 85, 102, 128, 151, 159, 174, 234, 255, 260, 279, 286, 288, 293, 306, 311, 313, 323]) explore diverse methods, including hyper-dimensional learning, adversarial training, model tuning, and more.

Adversarial Training and Defense Mechanisms: These studies focus on enhancing resilience to adversarial attacks through manifold-based adversarial training. Approaches such as GAN-based training [151] for adversarial robustness, ARMOR framework [40, 293] in computer vision, and ensemble learning [306] for Byzantine attack defense exemplify this category. Reference [313] uses a client update mechanism to enhance robustness by maximizing bias and variance during updates for adversarial defense, while [128] implements adversarial training with logits calibration (FedALC) to strengthen defenses against adversarial attacks in FL. **Backdoor and Model Poisoning Defense:** Strategies in this category aim to protect FL models from backdoors and poisoning attacks. Backdoor attack mitigation [234] uses client-side model tuning, while generative adversarial approaches for edge cases [40] target specific backdoor scenarios. Additionally, personalized training [47] and label correction networks [288] address label noise and poisoning attacks in domains such as medical imaging. Reference [323] uses a data-sharing strategy and attention-augmented model to obtain pre-trained model parameters and uniform data. Reference [174] investigates the impact of data poisoning attacks on model robustness using the FedAvg approach, highlighting FL's vulnerability to such threats. Reference [279] focuses on understanding how data heterogeneity influences the effectiveness of backdoor attacks in FL, aiming to quantify its impact on model performance. Expanding on backdoor defense, Reference [255] provides both theoretical and experimental insights into robustness against limited-magnitude backdoor attacks, establishing foundational guarantees for model safety. Last, Reference [260] introduces a backdoor defense approach using graph neural networks with subgraph trigger injection in FL for drug discovery, targeting domain-specific backdoor vulnerabilities.

Optimization and Adaptation for Robustness: Methods that focus on model tuning and dynamic adaptation to enhance FL robustness fall into this category. Studies like adaptive learning rates [311] adjust based on model behavior, while autonomous rate adjustments [286] manage learning progress. Hyper-dimensional learning [23] also speeds up training and mitigates communication costs, contributing to FL's robustness, while Reference [72] integrates teaching and learning for robust prediction, focusing on managing sensor noise and environmental perturbations through reliable data instances. **Privacy and Secure Aggregation:** Techniques in this category aim to ensure secure, private aggregation to counter-poisoning attacks. Studies such as encrypted verification schemes [85], SAFEFL with MPC [54], and contrastive clustering [102] for non-IID resilience, aims to safeguard aggregation processes against privacy breaches and malicious data manipulation. **Noisy Label Management and Resilience:** Several studies address the impact of noisy or heterogeneous data labels, crucial for robust FL model performance. Self-paced learning [81] mitigates label noise in medical imaging, while detection and exclusion of noisy federated weights [159] manage adverse data impacts. These approaches help improve model stability and performance in real-world FL scenarios. **Decentralized Data Integrity:** Some manifold methods involve decentralized data integrity strategies and enhance robustness against central model poisoning. Studies include FLGuard [22], which incorporates decentralized orchestration to withstand attacks and promote data fidelity. These manifold methods enhance FL's resilience against adversarial attacks, data diversity, and communication challenges, as summarized in Tables 9 and 10.

5.2 Server-side Treatment

5.2.1 Client Selection. Client selection in FL encompasses various innovative strategies across 83 studies to strategically choose clients, ensuring robust global model training while mitigating the impact of noisy or malicious contributions.

Approaches Utilizing Client Features and Behavior: Studies like Reference [181] focus on managing delays by categorizing models by origin, selecting representatives based on delays, and weighting contributions to ensure consistent participation. Secure aggregation via model filtering, such as the multi-Krum method with homomorphic encryption in Reference [238], and layer-wise network parameter distance approaches in Reference [147], which emphasize standard-range distances, are employed to mitigate offline client interference. Reference [88] ensures robust participation by calculating client contributions and aggregating data label-wise, while Reference [119] introduces automatic update clipping to improve clustering schemes based on vector norms. Filtering out malicious clients through univariate outlier detection, as seen in Reference [201], is another key method. Recent contributions include history-based anomaly detection [104], where adaptive trust scaling mitigates on-off, bad-mouthing, and good-mouthing attacks. Furthermore, in Reference [225], cosine similarity-based detection is integrated with personalized logit adjustment to handle non-IID data. Reference [140] addresses information leakage by utilizing population distribution for client selection, while Reference [17] employs trust bootstrapping based on a root dataset to weigh client updates for global aggregation. Reference [69] tracks the history of gradient updates with similarity metrics to detect unreliable clients, and Reference [4] uses clustering with cosine similarity and meta-learning to group and select high-quality clients. Last, Reference [251] dynamically extracts features and clusters clients based on sparse representations to exclude malicious clients, particularly in medical imaging applications. Reference [308] embeds watermarks in global models, tracking watermark recession to identify malicious clients.

Advanced Filtering and Aggregation Techniques: A variety of distance-based, clustering, and gradient-filtering approaches are utilized to secure robust client selection. For example, Euclidean distance calculations are used in Reference [67] to select benign nodes, while Reference [259] leverages norm-based gradient filters and trimmed-mean aggregation to filter out anomalous updates. The three-stage method in Reference [239] focuses on data complexity features to combat poisoning, while Reference [130] introduces binary permutation for secure aggregation, creating supernodes for enhanced synchronization. In studies such as References [145] and [202], multi-stage approaches emphasize pre-aggregation filtering and robust aggregation using the IOWA operator, respectively, to weigh model updates. Newer methods such as the dynamic gradient filter in Reference [34] and GradScore-based detection in Reference [266] have shown high resilience to Byzantine failures and targeted poisoning attacks. Additionally, reputation-based selection in Reference [179] and trust synthesizing mechanisms [55] use client trust scores dynamically, adapting client selection based on performance and historical reliability. Reference [166] integrates anomaly detection to filter out malicious updates but having potential high communication and computation overhead; scalability concerns. Reference [227] uses PCA-based reduction to filter malicious gradients by analyzing variations in local updates, and Reference [222] employs GAN-based countermeasures to detect and exclude anomalous client updates. References [300] and [257] implement mechanisms that use model-level unsupervised learning and truth discovery, respectively, to filter poisoned or malicious updates. References [144] and [235] apply specific testing datasets and adaptive clipping to detect and exclude malicious contributions. Reference [301] employs a double-aggregation mechanism with weighted averaging to further refine anomaly detection during model updates. Reference [132] employs voting mechanism for filtering abnormal directions and scaling mechanism for adjusting abnormal magnitudes but faces difficulty in converging the global model under high non-IID settings; potential vulnerability to adaptive attacks.

Client Selection Based on Resource Parameters: Optimal client selection based on resource metrics, such as computation power, data volume, and network bandwidth, is discussed in studies like Reference [70]. Trust scores and historical performance are also leveraged, as seen in Reference [295], where updates are forecasted based on previous iterations. Byzantine-robust methods, including zero-knowledge proof filtering in Reference [157], validate updates without data exposure, while cosine similarity metrics in Reference [163] focus on identifying malicious gradient contributions. Recent advancements further incorporate adaptive and dynamic reputation-based protocols [95] and hypermesh-based client grouping with secret sharing [154], offering refined client selection that scales across complex and high-dimensional environments. Reference [7] is focused on selecting clients based on resource parameters. It applies a sampling-based successive convex approximation to overcome non-convex challenges in training, addressing resource constraints by managing communication and computational demands within client selection. **Handling Channel Uncertainty and Complex Objective Optimization:** Channel reliability is vital in FL, particularly in noisy communication channels. To address this, Reference [52] introduces a method optimizing transmit power and beamforming to minimize MSE under channel uncertainty. Recent methods, including the Wasserstein distance-based approach in Reference [146] and Grad-Score for gradient monitoring [100], focus on optimizing the robustness of the aggregated model against high channel variability. The use of PCA for dimensionality reduction [167] also addresses channel uncertainty by clustering updates, while incentive mechanisms based on credit scores [63] balance computational efficiency and model fidelity. Reference [42] adjusts local updates with preference values, enhancing client selection by assessing loss reduction and historical performance, effectively optimizing for robust aggregation despite complex channel conditions.

Sophisticated Attack Mitigation and Robust Model Parameter Selection: To enhance robustness against sophisticated attacks, multiple methods employ anomaly detection and cosine similarity filtering. For example, Reference [272] integrates adaptive filtering with cosine similarity and Euclidean distance to combat Byzantine and label-flipping attacks, dynamically weighing client updates. In Reference [312], feature center separation techniques alongside outlier detection are employed to isolate malicious updates. Studies such as Reference [271] further add secure enclave-aided aggregation to protect updates, while Reference [169] uses saliency maps and structural similarity to detect gradient anomalies. Reference [36] leverages 3PC to securely select clients and aggregate models, enhancing privacy under Byzantine and poisoning attacks. References [134] and [68] use cosine similarity and element-wise sign functions, respectively, to identify malicious clients, while Reference [242] combines SVD and clustering to filter malicious updates. Additional methods such as References [57, 138, 173] incorporate mechanisms such as amplification, cosine distance, and adaptive clipping for selective filtering and punishment of malicious contributions, effectively enhancing robust parameter aggregation. Reference [296] estimates data divergence among clients to detect backdoor clients but the defense performance may degrade under non-IID settings and when adaptive attacks are specifically designed against the proposed method. Reference [115] uses private evaluations and clustering to detect outliers, and Reference [61] relies on VAE-based anomaly detection to exclude malicious updates.

Emerging Techniques for Aggregation Robustness and Incentivized Participation: Some methods are dedicated to robust aggregation with advanced filtering. For instance, robust federated clustering with meta-learning [27] aggregates models from similar clusters to minimize gradient divergence. Blockchain-based models, like Reference [229], introduce decentralized committees for model aggregation, and Reference [319] employs Bayesian frameworks to estimate client utility adaptively. Incentive-based schemes, such as References [58] and [248], promote fairness in client selection by rewarding trustworthy clients and mitigating label noise heterogeneity. These strategies ensure long-term participation and maintain trustworthiness. Reference [183]

uses Shapley values to measure client contributions, incentivizing participation by selecting valuable clients. Reference [298] introduces parameter compression and cosine similarity-based quality checks to select reliable clients, while Reference [249] combines gradient similarity with contract theory for fair incentive distribution among participants. Reference [232] applies similarity-based and spectral outlier detection in a four-pronged defense approach, and Reference [244] uses counters and authenticators to detect and block malicious clients in real-time, focusing on incentivizing and maintaining reliable client participation.

Aggregation Using Advanced Mathematical Models and Learning Techniques: New mathematical approaches, including Isolation Forest anomaly detection [240] and Hyperdimensional Computing-based encoding [91], are instrumental in client aggregation. These methods aim to reduce noise by identifying malicious models, while hyperdimensional encoding leverages cosine similarity for effective aggregation. Additionally, Reference [30] introduces Gradient Recycling, an approach combining reliable client selection with aggregation through gradient recycling to reinforce model consistency. Methods like Reference [210] use Siamese networks for CNN-based anomaly detection, while graph-theoretic correlation analysis [196] identifies backdoor attacks by observing client update correlations. Reference [302] applies reverse engineering with global reverse trigger creation to detect backdoor attacks through multi-step analysis. Reference [33] utilizes **Robust and Dynamic Gradient Filtering (RDGF)** with distance-based scoring and geometric median aggregation. Reference [264] combines clustering with cosine similarity to cluster models and prevents poisoning attacks. Reference [88] employs blockchain-based PoIS consensus to assess client reputation, and Reference [284] integrates cosine similarity-based aggregation with Lagrange coded computing. Reference [14] uses a two-stage approach for aggregation, estimating robust subspaces and filtering outliers, while Reference [25] employs an attention mechanism to weigh gradients for attack resilience. Finally, Reference [292] implements adversarial gradient regularization to optimize model performance while evading detection by aggregation robustness techniques. To this end, we provide a comprehensive comparative unit analysis indicating the potential gaps of these approaches in Tables 11–16.

5.2.2 Aggregation Approach Modification. The development of new aggregation algorithms is a novel approach to enhancing the robustness of FL by moving beyond traditional arithmetic averaging. These innovative algorithms aim to improve model convergence, accuracy, and security by aggregating client updates in a more adaptive and resilient manner. In the reviewed literature, 37 studies focus on this theme, each introducing unique aggregation techniques to address various FL challenges.

Robust Statistical Aggregation Methods: Robust statistical techniques improve the aggregation process by limiting the influence of outliers and malicious updates. For instance, geometric-median-based aggregation [117] and coordinate-wise median and trimmed mean [142] enhance resilience by isolating benign updates from noisy or faulty contributions. M-estimation aggregation [291] combines proximal algorithms with robust statistical metrics to handle high-dimensional data, while correntropy-based aggregation [150] applies maximum correntropy criteria, focusing on trusted updates in non-IID environments. The **Bootstrap Median-of-Means (bMOM)** [256] also provides outlier resistance through bootstrapping, aggregating updates effectively by ignoring extreme deviations. Reference [274] proposes using median-based and trimmed-mean-based gradient descent, providing resilience to Byzantine failures by mitigating the influence of extreme updates. Reference [228] adjusts client model weights based on probability distributions, penalizing clients with abnormal values to create a robust average. References [162] and [187] both utilize the geometric median; Reference [162] uses it to minimize weighted Euclidean distances between model updates, while Reference [187] iteratively applies a geometric median oracle to refine

robustness. Last, Reference [231] implements a node-wise trimmed mean by discarding extreme weight values from each local model, reducing the potential impact of outliers in aggregation.

Anomaly Detection and Filtering-based Aggregation: Anomaly detection plays a crucial role in identifying and removing malicious updates. Techniques such as **quadratic voting (QV)** [32] apply penalties to abnormal updates, while Selective Aggregation with **Conditional Variational Autoencoders (CVAEs)** [24] uses synthetic data validation to discard anomalous updates. Two-stage aggregation methods like in Reference [254] combine gradient similarity with noise injection, filtering out collusive behavior. Studies like confidence score-based scaling [41] adjust aggregation weights based on gradient reliability, while reputation-based detection and DBSCAN clustering in Reference [273] cluster benign clients, filtering outliers for robust model integrity. Reference [223] introduces a truth inference framework that estimates client reliability based on historical parameters to identify and exclude outliers. Reference [160] uses temporal divergence calculations to filter malicious updates by evaluating the historical behavior of client models. Reference [303] proposes a client sampling scheme using Euclidean distances to estimate adversary likelihood, filtering out abnormal updates through coordinate-wise median. Reference [314] combines multiple robust estimators, including NO-REGRET, FILTERING, and GAN-based algorithms, which rely on bucketing and robust estimation subroutines to detect and exclude Byzantine updates.

Reputation-driven and Adaptive Weighting Mechanisms: Reputation-based approaches assign weights dynamically based on client trustworthiness. Bayesian reputation-based aggregation [106] adjusts client influence using reputation scores derived from prior performance. Similarly, FedTop [233] weights client updates based on their ranking and contributions, while FedNAT [236] uses Wasserstein distance scoring to reward high-performing clients. The Layered-Switch aggregation in Reference [168] adds further robustness by randomizing weights across aggregation layers, increasing the difficulty of adversarial targeting. Reference [262] integrates robust normal aggregation with probabilistic perturbations and personalized model fusion, adapting weights based on individual client performance. Reference [105] combines robust mean-based aggregation with decentralized anomaly detection, adapting weights to counteract attacks such as poisoning and label-flipping. Reference [280] enhances Byzantine resilience by using secure cosine similarity and incentive-based weighting, rewarding clients who contribute high-quality, trustworthy updates.

Byzantine-resilient Aggregation: Byzantine-resilient methods protect against coordinated attacks by ensuring that only non-malicious updates are included in the aggregation. Median-based and trimmed-mean-based defenses [199] offload Byzantine filtering to network switches, increasing efficiency in handling large traffic volumes. The Krum approach [15] selects a single update as the most representative for the group, reducing the risk of contamination from malicious updates. Additionally, Convolutional Kernel Angle-based Defense Aggregation in Reference [320] applies cosine similarity to detect malicious gradients, making it effective for high-dimensional models under attack. Reference [135] uses a trimmed padding mean to calculate the median of parameters, minimizing the influence of Byzantine clients by averaging only benign client updates. Reference [203] compares Byzantine-tolerant methods such as Krum, Multi-Krum, RFA, and Norm-Difference Clipping to handle backdoor model poisoning. Reference [90] applies the **FL Trusted Coordinates (FLTC)** approach, selecting trusted coordinates based on direction and magnitude to defend against Byzantine updates. Reference [50] explores the use of FedAvg, FedProx, and Scaffold in non-IID settings, applying robust techniques for adversarial and Byzantine resilience.

Probabilistic and Randomized Aggregation Techniques: Probabilistic and randomized approaches add diversity to aggregation, reducing the likelihood of adversarial influence. Random noise and gradient filtering [254] use noise injection to prevent malicious clients from colluding.

Methods such as **Auto-Weighted Geometric Median (AutoGM)** [116] use probabilistic gradient assignments to balance between global and personalized models. FedNAT [236] integrates attention refinement and probabilistic scoring, enhancing update validity through random perturbations. Reference [206] introduces Zeno and coordinate-wise median aggregation, where clients are probabilistically ranked based on descent predictions in loss function, and model weights are combined by median for robustness. **Historical Information and Memory-based Aggregation:** Historical data is leveraged in some studies to identify and exclude unreliable updates based on past trends. Historical Information-based Recovery in Reference [19] estimates client updates from previous contributions, allowing for periodic correction and stability checks. Similarly, Weiszfeld aggregation [218] applies memory-based tuning to adapt aggregation thresholds based on historical gradient patterns, providing resilience over successive FL rounds. Reference [160] uses temporal divergence and client history to adjust weights dynamically, filtering out suspicious updates and minimizing poisoning impact. Reference [105] integrates anomaly detection with a memory-based mechanism, creating a decentralized record to support robust mean aggregation even in the presence of Byzantine clients.

Non-linear and Advanced Mathematical Models for Aggregation: More advanced mathematical approaches are adopted to process gradients in non-linear or complex forms. Selective **Maximum Correntropy Aggregation (MCA)** [150] applies fixed-point iteration for optimal aggregation, while Enhanced Weiszfeld aggregation [218] uses CNNs to refine feature selection from image data. bMOM [256] and attention refinement with Wasserstein distance [236] further enhance aggregation by combining advanced non-linear estimations and selective weighting. Reference [2] uses the Hessian matrix as a scaling mechanism, incorporating it into the aggregation process to handle data quality variations across models. This approach, inspired by Quasi-Newton methods, ensures robustness by adapting the aggregation process to different data qualities, contributing to a more precise model aggregation. We provide a comprehensive comparative unit analysis indicating the potential gaps of these approaches in Tables 17–19.

5.2.3 Aggregation Hyperparameter Tuning. The theme of hyperparameter tuning during model aggregation in FL encompasses 52 studies, each introducing unique techniques to optimize the aggregation process, enhancing FL's robustness. These approaches ensure that contributions from different clients are combined optimally, considering factors such as update quality, client reliability, and resistance to noise or adversarial attacks.

Trust and Reputation-based Tuning: Trust-based mechanisms assign weights to clients based on historical reliability or contributions. For instance, trust scores in Reference [237] are estimated from discrepancies in penultimate layer representations, while reputation scores in Reference [267] rely on a Bayesian framework for client utility estimation. Studies like FLEvaluate [62] apply fuzzy evaluation for trust-based aggregation, while ShapleyFL [219] uses Shapley value calculations to weigh client contributions adaptively. Trust-based model updates are refined in FedValidate [321], which validates client credibility before aggregation. Additionally, trustworthy model updates [318] leverage local and global update evaluations in industrial cyber-physical systems, addressing free-riders and poisoning attacks. Reference [11] uses cosine similarity to gauge client alignment, adjusting learning rates and calculating reputation scores to select reliable local models. Reference [118] implements an auto-weighted FL approach that adjusts client weights according to empirical risk, penalizing clients with low performance, while Reference [237] estimates trust scores based on penultimate layer representations, emphasizing alignment with trustworthy clients to ensure robust updates in aggregation. Reference [208] uses reputation-based weighting of gradients calculation and uses flip-score to detect malicious clients, but this work has dependency on the assumption that gradient flips are rare in benign conditions and may not cover scenarios where benign clients exhibit similar behavior.

Weighted and Dynamic Aggregation: Studies focusing on weighted aggregation use dynamic, context-sensitive weights to improve robustness. For instance, contribution-wise weighted aggregation in Reference [125] assigns scores based on client contributions and uses a trimmed-normalized aggregation rule to counter scaling attacks. Cosine similarity-based weighting in Reference [139] measures both Euclidean distance and cosine similarity, refining client selection. Weighted aggregation using final-layer adjustments [270] identifies malicious models by embedding backdoor triggers, while FL-GQM [139] uses distributed scoring to increase resistance to adversarial clients. Reference [108] prioritizes clients with higher rewards, reducing cooperation with lower-reward clients for network efficiency. Reference [110] applies Shapley values to measure sample contributions, re-weighting updates based on individual data relevance, while Reference [96] leverages k -reliable neighbors to obtain clean and similar data for reliable training, dynamically adjusting weights based on data quality and relevance. Reference [114] classifies DNN parameters into critical and noncritical, applies different update rules, and uses weighted aggregation based on learning efficiency but has potential dependency on accurate classification of parameters and may not perform well in cases with extreme noise. Reference [309] utilizes existing robust aggregation rules like trimmed mean. Reference [83] uses hierarchical fusion of heterogeneous models to mitigate noise; they employ a heteroArc fusion layer for aggregating heterogeneous models.

Clustering and Group-based Hyperparameter Tuning: Several studies use clustering to separate benign from malicious updates. **Agglomerative Hierarchical Clustering (AHC)** in Reference [122] leverages clustering techniques to enhance Byzantine robustness. Mini-FL [129] clusters clients by non-IID features (e.g., geographic, temporal) for grouped model aggregation, while two-stage clustering and screening in Reference [124] uses centerpoint aggregation, combining clustering with discrete geometry to handle large-scale models. Clustering-based Byzantine-robust aggregation is further explored in Reference [175], which applies coordinate-wise screening alongside centerpoint aggregation. Reference [20] clusters clients into groups for group-wise aggregation, training separate global models and using majority voting to classify test inputs. Reference [112] uses metrics learning combined with unsupervised clustering (K-means) to detect and filter malicious data based on learned representations. Similarly, Reference [277] clusters model parameters, performing robust aggregation within each cluster to ensure resilience against Byzantine and adversarial attacks. Reference [60] determines reconstruction errors of updates using a variational autoencoder, discarding updates with excessive errors. **Game-theoretic and Anomaly Detection Tuning:** Game-theoretic approaches identify and discard unreliable clients. For instance, game-theoretic filtering in Reference [221] detects malicious updates, while anomaly scores in Reference [281] and RONI/TRIM in Reference [45] identify updates with high error rates or inconsistencies. Homomorphic encryption and secret sharing [71] strengthen the privacy and robustness of client updates, while Moreau envelope smoothing [278] mitigates noisy updates using an envelope-smoothed approximation. Reference [131] employs spatial-temporal pattern analysis to detect anomalous gradients, filtering them through cosine similarity and clustering. Reference [77] uses isolated forest-based anomaly detection by building decision trees and assigning anomaly scores to client updates for malicious update trimming. Reference [193] applies a supervised anomaly detector to filter harmful updates, distinguishing benign from malicious updates to ensure a cleaner model aggregation process.

Privacy-enhanced Tuning Using Encryption: Encryption-based tuning strategies provide robustness against attacks by obscuring updates. For example, distributed Paillier encryption with zero-knowledge proofs [155] ensures privacy and Byzantine resistance. Studies like Mask-NFRT [71] employ secret sharing for dropout tolerance, while double-masking techniques [149] secure client updates with pseudo-random generator-based encryption. Privacy is further enhanced in differential privacy frameworks such as DPFA and DPLoss [51], which implement differential privacy

within aggregation. Reference [53] uses a verification dataset and confidence scores to identify malicious clients, enhancing aggregation security while requiring a private shared dataset. Reference [268] maintains uniform decision boundaries by exchanging class-wise centroids between clients and the server, synchronizing local model features for consistency despite noisy labels, though this approach requires sharing some client data features with the server. **Trust Metrics and Consensus Mechanisms:** Some approaches rely on trust metrics and consensus to filter malicious updates. Consensus mechanisms in Reference [18] align global models based on majority voting, and alarm-based anomaly detectors [64] detect poisoned updates for exclusion. Client trust and consensus with geometric median [38] uses Hamming distance with a clean dataset to bootstrap trust scores and support secure aggregation. Reference [188] evaluates client reliability by assessing parameter alignment with the server's values, allowing the exclusion of clients with weak alignment. Reference [214] uses historical Euclidean distances to differentiate between benign and malicious updates, assuming that benign clients have more consistent update distances, which guides the server in selecting trustworthy updates.

Adaptive and Sensitivity-based Tuning: Adaptive methods adjust model hyperparameters based on sensitivity to attack parameters or specific client behaviors. For example, adaptive divergence scoring [5] uses divergence and outlier scores to adjust weighted averaging. Sensitivity analysis-based Krum [99] estimates attacker count to adapt Krum's aggregation approach, ensuring robust aggregation even with Byzantine adversaries. Reliability-based weighting [43] applies a mixture of experts to score updates adaptively based on model quality. Reference [212] adapts privacy budgets based on Wasserstein distance, generating uncertainty sets that account for individual client variability and optimizing model performance. Reference [207] uses a tracking-based stochastic approximation to offset noise impacts in over-the-air transmissions, adapting the model for robust convergence under noisy channel conditions, particularly useful in decentralized networks with transmission constraints.

Cryptographic Protocols for Tuning: Cryptographic protocols support robust tuning in high-security FL setups. Additive Secret Sharing and Oblivious Transfer in Reference [149] leverage cryptographic measures for aggregation, while Zero-knowledge Proofs with Paillier encryption [155] ensure privacy and defend against Byzantine attacks. Federated Aggregation with Homomorphic Encryption [126] applies encryption to protect against model poisoning attacks, particularly for non-IID data setups. Reference [26] employs hierarchical aggregation with sub-aggregators, where users send split parameters to sub-aggregators before the final consolidation, boosting privacy in FL. Reference [241] addresses communication quality by jointly allocating bandwidth and quantization bits, reducing transmission errors and ensuring reliable updates. Reference [216] assigns transmission time slots to client groups and applies robust aggregation to mitigate Byzantine attacks, promoting secure aggregation across heterogeneous communication channels. To summarize, we provide a comprehensive comparative unit analysis indicating the potential gaps of these approaches in Tables 20–23.

6 Open Issues and Potential Solutions

This section identifies key challenges and outlines potential solutions in robust FL. The aim is to address the reviewer's suggestion by offering detailed insights and deriving new directions based on the systematic analysis in this survey, incorporating specific issues, solutions, and technical details.

6.1 Advancing Robust and Secure Model Aggregation Algorithms

Traditional aggregation techniques like FedAvg inadequately handle the multifaceted challenges of data heterogeneity, malicious updates, and adversarial attacks in FL. Current solutions often

specialize in addressing isolated issues; for instance, robust aggregation methods such as Krum and Trimmed Mean focus on Byzantine robustness but may fail under non-IID data distributions. To provide comprehensive security, advanced aggregation algorithms must incorporate multi-dimensional strategies. One promising direction is the development of adaptive weighting schemes that dynamically adjust client contributions based on trust metrics derived from client behavior, such as update consistency, loss convergence patterns, or model divergence. For example, methods like coordinate-wise median aggregation can mitigate the impact of outliers but may be computationally intensive in high-dimensional models. Integrating anomaly detection mechanisms into the aggregation process can help identify and exclude malicious updates. Future work should focus on scalable algorithms that minimize computational overhead while effectively balancing robustness and accuracy, possibly through techniques such as gradient sparsification or dimensionality reduction in the aggregation process. Additionally, **secure multi-party computation (SMPC)** protocols can be employed to securely compute aggregates without exposing individual updates, enhancing both security and privacy.

6.2 Integrating Hybrid Server-client Defense Mechanisms

The current focus on either server-side (e.g., robust aggregation) or client-side (e.g., local differential privacy) defenses in FL creates vulnerabilities exploitable by sophisticated attackers who can circumvent isolated defenses. Hybrid approaches that integrate both server and client perspectives are critical for establishing a resilient FL framework. For instance, implementing secure aggregation protocols like Bonawitz et al.'s Secure Aggregation allows clients to jointly compute aggregates without revealing individual updates, while the server can apply anomaly detection on aggregated results using statistical tests or machine learning classifiers. Techniques like **Federated Adversarial Training (FAT)**, where both clients and the server participate in generating and defending against adversarial examples, can enhance robustness against distributed attacks. Additionally, feedback loops between the server and clients, where the server provides feedback on the quality or trustworthiness of client updates (e.g., through reputation scores or trust metrics), can enable clients to adjust their local training strategies dynamically. This co-design of detection algorithms leveraging both global and local information can improve the system's ability to identify and mitigate threats, such as backdoor attacks or model poisoning.

6.3 Harnessing Ensemble Learning Techniques for Federated Robustness

Ensemble learning can effectively address the diversity of challenges faced in FL by integrating multiple robust techniques into a cohesive framework. Beyond traditional majority voting or averaging, ensemble-based approaches in FL can leverage advanced methods like Federated Stacking, where a global meta-model learns from the outputs of diverse client models to enhance resilience against attacks. For example, each client could train a local model with different architectures or hyperparameters, and the server could combine their predictions using weighted ensembling based on performance metrics such as accuracy or loss. By combining diverse defense mechanisms tailored to attack types and data distributions, ensemble learning improves robustness while reducing reliance on any single strategy. Additionally, incorporating techniques like bagging and boosting in the federated context can help mitigate overfitting and improve generalization. The integration of model diversity metrics during aggregation, such as measuring cosine similarity or kernel alignment between updates, can further bolster FL defenses against adversarial attacks and heterogeneous data challenges, ensuring that the ensemble maintains high performance even under attack.

6.4 Developing Adaptive Mechanisms for Dynamic Threats in FL

FL systems operate in dynamic environments where threats evolve rapidly, and static defense mechanisms may struggle to adapt to such changes. Incorporating adaptive strategies that respond to observed behavior and performance metrics is essential. Adaptive learning rates for client training, adjusted based on local loss convergence or gradient variance, can help maintain stability and robustness. Techniques like **Reinforcement Learning (RL)** can be employed to dynamically adjust hyperparameters such as the number of local epochs, batch size, or regularization strength in response to changing threat levels. For example, if a client's updates deviate significantly from the global model, then the system could reduce that client's influence or increase scrutiny of its updates through adaptive weighting or outlier detection algorithms like **Local Outlier Factor (LOF)**. Adaptive client selection mechanisms can prioritize clients with more reliable updates or diverse data, improving the robustness and fairness of the aggregated model. Furthermore, implementing online anomaly detection algorithms that continuously monitor model updates for signs of poisoning or backdoor attacks can enable timely responses to evolving adversarial threats, utilizing techniques such as **Principal Component Analysis (PCA)** for dimensionality reduction and anomaly detection.

6.5 Designing Lightweight and Efficient Encryption Protocols for FL

Integrating encryption into FL to ensure security and privacy often introduces significant computational overhead, particularly in resource-constrained edge environments. To address this, lightweight symmetric encryption schemes like stream ciphers (e.g., ChaCha20, Salsa20) can be employed for encrypting model updates, reducing computational load compared to more heavyweight asymmetric schemes. Additionally, leveraging hardware acceleration available on modern devices (e.g., ARM processors with NEON SIMD instructions) can enhance cryptographic operations' efficiency. Exploring hybrid encryption schemes that combine symmetric encryption for data transmission and homomorphic encryption for secure computation can balance security and performance. For instance, partially homomorphic encryption schemes such as Paillier or Batched RSA can enable secure aggregation of encrypted model updates without decrypting them, although they may be limited to additive operations. Advancements in efficient SMPC protocols, such as Secret Sharing schemes (e.g., Shamir's Secret Sharing), can facilitate secure aggregation with lower computational overhead. The development of these lightweight encryption protocols is instrumental in making FL scalable and practical in real-world deployments where clients have limited computational resources and energy constraints.

6.6 Enhancing Real-time and Online Robustness in FL

FL's distributed nature and asynchronous operation make it susceptible to adversarial attacks and concept drift, especially in scenarios where data distributions change over time (non-stationary environments). To enhance real-time robustness, implementing asynchronous model aggregation schemes like FedAsync can reduce delays associated with synchronous updates and allow faster integration of client updates. However, asynchronous updates introduce challenges in maintaining model consistency and convergence, which can be addressed using staleness-aware aggregation methods that weigh updates based on their freshness. Hybrid adversarial detection mechanisms that combine real-time monitoring with predictive analytics can counter threats effectively. For example, real-time anomaly detection algorithms can flag suspicious updates based on statistical deviations using techniques like the **Sequential Probability Ratio Test (SPRT)**, while predictive models using **Recurrent Neural Networks (RNNs)** or **Long Short-Term Memory (LSTM)** networks anticipate potential attacks based on historical patterns. Incorporating online learning

methods, such as Online Gradient Descent or adaptive forgetting factors, can help models adapt to concept drift by giving more weight to recent data. Techniques like **Elastic Weight Consolidation (EWC)** can prevent catastrophic forgetting when adapting to new data distributions. These strategies improve robustness against evolving data distributions and adversarial patterns, ensuring that the FL system remains effective over time.

6.7 Balancing Performance, Security, and Fairness in FL

Tradeoffs between robustness, accuracy, and fairness remain a critical challenge in FL. For instance, employing strong differential privacy mechanisms enhances privacy but can degrade model accuracy due to added noise. To address this, multi-objective optimization frameworks that explicitly account for these tradeoffs are needed. Techniques like Pareto Optimality can help find solutions that balance competing objectives. RL can be used to dynamically adjust defense mechanisms and hyperparameters based on their impact on performance metrics. For example, an RL agent could learn to adjust the level of noise added for privacy based on the current model accuracy and fairness metrics. Personalized FL approaches, where models are tailored to individual client data distributions and requirements, can reduce disparities while maintaining overall robustness. **Federated Multi-Task Learning (FMTL)** allows different clients to have models customized to their specific tasks, improving fairness and potential robustness. Additionally, fairness-aware FL algorithms that incorporate fairness constraints into the optimization objective can help ensure equitable performance across clients. Techniques like Adaptive Weighting can balance the contributions from different clients based on data quality, contribution to model performance, and adherence to fairness criteria.

6.8 Evolving FL Architectures for Scalability and Efficiency

The traditional star-based architecture of FL, where a central server coordinates all clients, is increasingly inadequate for large-scale deployments with diverse clients due to communication bottlenecks and single points of failure. Decentralized architectures, such as peer-to-peer networks or gossip-based protocols, can reduce reliance on a central server and enhance scalability. For instance, **Decentralized FL (DFL)** allows clients to communicate and aggregate models with their neighbors in a network graph using algorithms such as Push-Sum or Gossip Averaging. Hierarchical FL architectures introduce intermediate aggregation layers, where edge servers or regional aggregators collect updates from nearby clients before passing them to the central server. This approach reduces communication overhead and can handle data heterogeneity more effectively by grouping clients with similar data distributions. Blockchain-enabled FL frameworks offer a novel approach to improving transparency and trustworthiness in data aggregation and transmission. Smart contracts can automate and secure the aggregation process, while consensus mechanisms such as **Proof of Stake (PoS)** or **Practical Byzantine Fault Tolerance (PBFT)** can ensure the integrity of updates. These architectural innovations are critical for adapting FL systems to complex and large-scale applications, but they also introduce new challenges in terms of latency, security, and protocol design that must be addressed.

6.9 Achieving Cross-domain Robustness in FL

Cross-domain FL faces significant challenges due to differences in data characteristics, feature spaces, label distributions, and security requirements across domains. Transfer learning techniques, such as fine-tuning a pre-trained model on new domain-specific data, can enhance cross-domain robustness. In FL, this could involve clients in a new domain adapting the global model by fine-tuning their local data while contributing to the global update. Meta-learning approaches, such as **Model-Agnostic Meta-Learning (MAML)**, where the model is trained to be

easily adaptable to new tasks, can be effective in enabling models to quickly adapt to new domains with minimal data. Joint learning frameworks that facilitate collaboration across domains without compromising privacy and security, such as **Federated Transfer Learning (FTL)**, represent a promising direction for future research. FTL can leverage knowledge from related domains even when there is little overlap in data features or labels. Techniques such as domain adaptation and adversarial learning can help align feature spaces across domains, improving the generalization of the global model. Additionally, incorporating multi-view learning or multi-modal learning can further enhance the model's ability to handle heterogeneous data from different domains.

6.10 Providing Theoretical Guarantees for Robust FL

Despite empirical advancements, robust FL lacks comprehensive theoretical foundations to validate its effectiveness under diverse scenarios. Rigorous definitions of robustness metrics, encompassing both data heterogeneity and adversarial resilience, are essential. Establishing theoretical bounds for attack resilience involves analyzing the convergence properties of FL algorithms in the presence of malicious clients and quantifying the maximum allowable proportion of adversaries for which convergence can still be guaranteed (Byzantine fault tolerance thresholds). Additionally, deriving convergence rates under non-IID data distributions and asynchronous updates is critical. Developing certified verification frameworks for robustness, akin to formal methods in software verification, can ensure reliable deployment of FL systems in critical applications. Techniques from robust statistics, such as influence functions and robustness measures (e.g., breakdown point, influence curve), can be adapted to analyze the robustness of FL algorithms. Formalizing privacy guarantees through differential privacy and understanding their impact on convergence and robustness is also important. Furthermore, applying probabilistic analysis and concentration inequalities (e.g., Hoeffding's inequality and Bernstein's inequality) can provide insights into the behavior of FL algorithms under uncertainty. These theoretical foundations will provide a quantitative benchmark for evaluating FL methods and will help in designing algorithms with provable security and performance guarantees, essential for critical applications such as healthcare and autonomous systems.

7 Conclusion

This article has presented a comprehensive and insightful exploration into the realm of robust FL through an SLR guided by the PRISMA framework. The review's meticulous analysis has unveiled a diverse array of approaches and methodologies dedicated to enhancing the resilience of FL models in the face of adversarial challenges, noise, and the complexities of decentralized data. The identified eight themes—ranging from innovative objective regularization and optimizer modification strategies to the implementation of differential privacy, client selection, new aggregation algorithms, and meticulous hyperparameter tuning during model aggregation—collectively depict the breadth and depth of strategies employed by researchers to ensure the robustness of FL. By synthesizing these themes, this review not only encapsulates the present state-of-the-art but also paves the way for the future of research in the domain.

This SLR also points towards promising directions for further investigation. It underscores the potential of hybrid models, ensemble methods, and adaptive mechanisms to bolster the resilience of FL systems even further. The convergence of these methods, combined with advancements in data privacy and security, has the potential to address the complex challenges posed by adversaries and heterogeneity in decentralized data sources. As the FL landscape evolves rapidly, this review stands as an invaluable resource, guiding researchers, practitioners, and policymakers toward developing robust FL solutions. The insights derived from this analysis can

serve as a foundational framework for the creation of more secure and reliable FL systems, ultimately shaping the future landscape of decentralized ML in the pursuit of privacy, security, and performance.

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Appendix

A Consolidated Thematic Result Analysis

A.1 Client-side Treatment

A.1.1 Objective Regularization. A comprehensive comparative analysis of the objective regularization-based approaches for ensuring the robustness of FL is provided in Table 5, which highlights potential gaps in their effectiveness as well.

Table 5. Objective Regularization-based Treatments to Attain Robustness in FL

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[120]	Robust to data and model poisoning attacks	L2 regularization-based training	Uses the existing L2 regularization, analyzed in the FedProx method	CV	Fashion MNIST, CelebA and FEMNIST	Experimental	No exploration on the applicability of the proposed method to other attacks such as backdoor attacks; Limited experiments performed in the classical FL context; No experiments on non-IID data
[66]	Robust to malicious updates	Jacobian regularization-based training	Uses Jacobian regularization to increase model stability against model perturbations	CV	CIFAR10, MNIST, SVHN, and Fashion-MNIST	Experimental	No treatment on the models aggregation; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[8]	Robust to noise from wireless communications	Sampling-based successive convex approximation-driven regularization	Uses a regularization for the loss function approximation via sampling-based successive convex approximation	CV	MNIST	Theoretical and Experimental	Regularization is not made adaptive; Limited experiments performed in classical FL context; No experiments on non-IID data
[27]	Robust to noisy labeled data	Loss function correction using a re-weighted mechanism	Edge devices learn to assign the local weights of loss functions in noisy labeled datasets and cooperate with the central server to update global weights	CV	MNIST, and CIFAR10	Theoretical and Experimental	Requires a local clean test dataset in each client; Limited experiments on non-IID data
[82]	Robust to noisy labels (label flipping attacks)	Local self-regularization method	Improves local training by preventing memorization of noisy labels and narrowing the gap between original and augmented data using self-distillation	CV	MNIST, Fashion-MNIST, CIFAR10, Clothing1M	Experimental	Requires augmented data; Limited experiments on non-IID data; Limited poisoning attacks tested
[322]	Robust to Byzantine attacks	Secure aggregation	RiseFL framework using probabilistic L2-norm checks and hybrid commitment schemes	Medical Imaging	Synthetic, Real-world tabular data, Medical image datasets	Experimental and Theoretical	High computation and communication cost; Scalability challenges with large numbers of clients
[101]	Robust to adversarial attacks and Byzantine attacks	Adaptive selection of loss function	Adaptive loss function selection based on whether a client is under attack, combined with Byzantine-robust aggregation	Intrusion Detection	CICMalDroid 2020	Experimental	The combination of adversarial training and Byzantine-robust aggregation did not complement each other as expected, leading to potential inefficiencies in model training stability and convergence
[305]	Robust to Byzantine attacks	Huber loss minimization	Minimizing multi-dimensional Huber loss to aggregate gradients	Not specified	Synthesized and real data	Theoretical and Experimental	Needs further analysis for extreme cases of non-IID data and highly imbalanced data

A.1.2 Optimizer Modification. The optimizer modification-based approaches for ensuring FL robustness are compared comprehensively in Table 6.

A.1.3 Differential Privacy Employment. We provide a comprehensive comparative unit analysis indicating the potential gaps in the differential privacy employment-based approaches for ensuring FL robustness in Table 7.

A.1.4 Additional Dataset Requirement and Decentralization Orchestration. We provide a comprehensive comparative analysis, outlining the potential gaps in the additional dataset requirement and decentralization orchestration-based approaches for ensuring FL robustness in Table 8.

A.1.5 Manifold. The manifold methods for ensuring FL robustness are summarized in Tables 9 and 10.

A.2 Server-side Treatment

A.2.1 Client Selection. We provide a comprehensive comparative unit analysis indicating the potential gaps of the client selection-based approaches for ensuring FL robustness in Tables 11–16.

A.2.2 Aggregation Approach Modification. We provide a comprehensive comparative unit analysis indicating the potential gaps of the aggregation approach modification-based approaches for ensuring FL robustness in Tables 17–19.

Table 6. Optimizer Modification-based Treatments to Attain Robustness in FL

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[200]	Robust to malicious clients	Modification of the SGD optimizer for making it Byzantine-resilient	Simplified convergence theory is used to develop the generic Byzantine-resilient SGD	-	-	Theoretical	No new robustness method proposed; No experimental validation
[137]	Robust to malicious models	Distributed gradient-descent with robust gradient aggregation	Allowing all the non-faulty agents to compute the minimum point of only the non-faulty agents' aggregate cost, instead of all the agents	-	-	Theoretical	No non-IID data consideration; Just model poisoning attacks tested without data poisoning attacks; Limited attack-robust baselines
[44]	Robust to Byzantine attack or noisy clients	SGD-based best effort voting to Byzantine attack	SGD is modified by allowing all the workers to send their local updates at their maximum transmit power to make it robust to the Byzantine attack	IoT	MNIST	Theoretical and Experimental	It performs better in strong Byzantine attack but in the weakest Byzantine attack, sometimes it fails. No experiments on non-IID data
[315]	Robust to poisonous models	Use of stochastic average gradient algorithm	Reducing the compression noise with gradient difference compression to improve the Byzantine-robustness	CV	COVTYPE	Theoretical and Experimental	No considering Non-IID settings; Limited Datasets (only one); Limited FL settings; Limited baselines; Limited attack settings (specifically fewer attack scenarios consideration)
[184]	Robust to Byzantine attacks	Stochastic average gradient algorithm (SAGA) for client-side training followed by resampling strategy	SAGA is designed to update the local model to eliminate the outer variation caused by Byzantine clients via resample-then-aggregate strategy for selecting the variation-reduced gradient to aggregate with geometric median aggregation method	CV	MNIST, COVTYPE, CIFAR100	Theoretical and Experimental	Limited non-IID data settings; Unstable performance on non-IID data with diverse attacks
[261]	Robust to Byzantine nodes	Gradient-similarity-based secure consensus	Each consensus round involves all nodes transmitting the IDO sub-gradients, which are carefully clustered and divided. The dynamic removal of Byzantine gradients is then carried out in each round of the consensus procedure.	CV	MNIST and EMNIST	Experimental	Increases computation and communication costs, Limited experiments performed in the classical FL context; No experiments on non-IID data
[198]	Robust to Affine distribution shifts	A minimax optimization method	A minimax distributed optimization method is formulated to guard against affine distribution shifts	CV	MNIST, and CIFAR10	Theoretical and Experimental	Limited affine distribution shifts tested; Limited experiments performed in classical FL context; No experiments on non-IID data
[265]	Robust to targeted and untargeted attacks	Robust Ternary Gradients Aggregation combining ternary gradients quantization	Ternary quantization compresses client gradients to 2 bits per vector point. Error-feedback gauges real-compressed gradient variance. The server's aggregation detects malicious clients based on risk scores from divergence optimization.	CV	MNIST, and CIFAR10	Theoretical and Experimental	Exploring the relationship between quantization methods and robust aggregation algorithms are missing; Limited experiments performed in classical FL context; Limited experiments on non-IID data
[269]	Robust to Byzantine attacks and poisoning attacks	Buffered asynchronous stochastic gradient descent	Use of buffers to store gradients before aggregation, adaptive reassignment of workers to buffers, local momentum for enhanced Byzantine resilience	CV	CIFAR-10, ImageNet	Experimental and Theoretical	Potential delay in convergence due to buffer reassignment and asynchronous updates; requires fine-tuning of buffer size and reassignment interval for optimal performance
[165]	Robust to inference attacks and poisoning attacks	Polymorphic encryption for securing data exchanges	Uses AES-256 with polymorphic keys for encrypting each message exchange. Gradient Descent for optimization	Medical Imaging	Simulated dataset, SHAREEDB, and Surgical binary dataset	Experimental	Limited to SVM, high computational cost due to encryption, challenges in scalability with large client networks
[230]	Robust to model poisoning attacks and Byzantine attacks	Adaptive model update	Adaptive tuning of local epochs, step size adjustment, and robust aggregation by supplanting outlier updates	CV and Text Classification	MNIST, CIFAR-10, Shakespeare	Experimental and Theoretical	Performance may degrade in high attack scenarios; needs further exploration of efficiency in large-scale models
[211]	Robust to noisy labels	Robust Optimization	Utilizes a Wasserstein distance-based certified robust region to normalize noisy labeled data, combining it with clean data for training	Intrusion detection	ISCXVPN-2016	Experimental and Theoretical	The method introduces additional computational overhead due to the DRO process and the need to estimate global distribution in each round
[204]	Robust to adversarial data	Client selection	Wasserstein distance-based ambiguity set for DRO, Stochastic Proximal Gradient Descent (SPGD) algorithm	CV	MNIST, Fashion MNIST, CIFAR-10, CIFAR-100	Experimental and Theoretical	High computational complexity in large-scale settings, need for careful tuning of hyperparameters, potential challenges in non-IID data scenarios
[224]	Robust to Byzantine attacks	Byzantine-resilient distributed gradient descent	Use of coordinate-wise trimmed mean aggregation and gradient compression techniques to handle heavy-tailed data	Text Classification	Synthetic datasets, and Boston Housing Price dataset	Experimental and Theoretical	Challenges in scalability due to the heavy computational and communication costs, particularly when dealing with large-scale networks and datasets

A.2.3 Aggregation Hyperparameter Tuning. We provide a comprehensive comparative unit analysis indicating the potential gaps of the aggregation hyperparameter tuning-based approaches to ensure the robustness of FL in Tables 20–23.

Table 7. Differential Privacy Employment-based Treatments during Model Training to Attain Robustness in FL

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[290]	Robust against poisoning attacks	Differential privacy	Joint differential privacy-based FL for limiting the influence of adversaries	CV	MNIST, CIFAR10	Theoretical and Experimental	No consideration of model poisoning attacks (just limited data poisoning attacks); No non-IID consideration; No attack-robust baselines
[220]	Robust to backdoor attacks (specifically targeted model update poisoning attacks)	Differential privacy	Defenses contain weak differential privacy for model update attacks including unconstrained and constrained attacks for fixed-frequency attacks and random sampling	CV	EMNIST	Experimental	Just robustness evaluation in FL with existing backdoor attacks and defense; Evaluation on one dataset, EMNIST only; Limited number of defense methods; Limited backdoor attacks considered; No Non-IID setting considered
[148]	Robust to malicious updates	Differential Privacy followed by random sub gossip update and incentive-weighted aggregation	It encrypts local updates with Local DP (LDP), uses random gossip updates to thwart malice and selects low mean absolute error local models for global aggregation-	Text Classification	Reuters, Newsgroups, and Ohsumed	Experimental	No consideration of non-IID data; Limited FL hyperparameters settings; No comparison baselines; Just adapt to asynchronous FL
[156]	Robust to Byzantine adversarial attacks	Differentially private summation protocol	It utilizes a shuffle protocol for summation to realize the DP encryption for exchanging model parameters and global aggregation and uses existing Byzantine-robust stochastic aggregation to aggregate the global model	CV	MNIST, Fashion-MNIST, and CIFAR10	Theoretical and Experimental	Limited Byzantine attacks considered; Limited FL hyperparameters settings; Limited comparison baselines
[79]	Robust to model poisoning attacks	Differential privacy and Gaussian noise	VPPFL scheme with DP-based perturbation and verification mechanism	CV	MNIST, CIFAR-10	Experimental, Theoretical	Overhead due to DP noise, scalability for large user groups
[65]	Robust to Byzantine attacks	DP-based training	SIREN+ framework with local DP and proactive alarming based on model accuracy	CV	Fashion-MNIST, CIFAR-10	Experimental	Potential communication overhead and computation burden due to proactive alarming and DP mechanisms
[190]	Robust to adversarial attacks	Adversarial training	FedAIC framework using adversarial training (AT) with logits calibration for balancing class distributions	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	Limited performance on extremely unbalanced datasets; potential overhead due to adversarial training
[290]	Robust to model poisoning attacks and malicious clients	Differential privacy	Joint differential privacy to motivate honest participation and ensure robustness	CV	MNIST, CIFAR-10, LFW	Theoretical and Experimental	Potential issues with the practical implementation of game-theoretic mechanisms
[176]	Robust to Byzantine attacks and poisoning attacks	DP-based training	The proposed algorithm combines Differential Privacy with Online Mirror Descent to ensure privacy and convergence in the presence of Byzantine clients	CV	MNIST, CIFAR-10	Both Experimental and Theoretical	Potential challenges in scalability due to the computational complexity of the proposed algorithm, especially in large-scale systems with many Byzantine clients and non-IID data scenarios
[297]	Robust to Byzantine attacks and poisoning attacks	DP-based training	Integrating sparsification and momentum-driven variance reduction with a client-level DP mechanism	CV and Text Classification	Fashion-MNIST, Shakespeare dataset	Both Experimental and Theoretical	High variance in the aggregated model update when dealing with large models; potential for increased computational complexity due to variance reduction techniques

Table 8. Additional Dataset and Decentralization Orchestration-based Treatments to Attain Robustness in FL

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[246]	Robust to poisoning attacks	Blockchain multi-layer decentralized FL framework	Blockchain is used to secure the privacy and data safety, provide resilience on poisoning attacks to the central model, prevent the central server failure or malfunction	IoT	MNIST	Experimental	Only one toy dataset (MNIST) considered; No new aggregation method proposed; No non-IID consideration; Limited FL hyperparameters; No attack-robust baselines; Limited poisoning attacks
[46]	Robust to noisy and heterogeneous clients	Utilizing public data, noise-tolerant loss function, and re-weighting scheme	Heterogeneous model communication employs a public dataset for feedback alignment. Robust loss mitigates internal label noise. A novel client confidence re-weighting handles challenging noisy feedback in collaborative learning.	CV	CIFAR10, and CIFAR100	Experimental	Requires a global public dataset; Limited experiments performed in classical FL context; limited experiments on non-IID data
[253]	Robust to Byzantine attacks	Reference dataset-based algorithm along with a practical Two-Filter algorithm (ToF)	Creates a reference dataset to examine the performance of the uploaded weights computed by each node from their private dataset using training losses	CV	MNIST, CIFAR10	Experimental	Requires a local test dataset in each client, Limited poisoning attacks tested; Non-IID dataset is not considered
[49]	Robust to model poisoning attacks and malicious and unreliable clients	Blockchain-enabled FL	FedAnil model: Combines clustering, blockchain technology, and encryption techniques to ensure robustness and privacy-preservation in FL	CV	Sent140, Fashion-MNIST, FEMNIST, CIFAR-10	Experimental	The model may not fully resist advanced label-flipping attacks and large-scale collude attacks; potential for divergence in certain non-IID settings
[16]	Robust to model poisoning attacks and Byzantine failures	Decentralized FL	Uses blockchain and role rotation	CV	CIFAR-10, PEMS-BAY	Experimental	Need for more comprehensive testing across different types of attacks.
[48]	Robust to Byzantine attacks	Byzantine-resilient consensus protocol	Decentralized Byzantine-resilient consensus using Smart Contracts and blockchain, Differential Privacy for data protection	CV	MNIST	Experimental	High computational cost for blockchain operations, potential issues with scalability in large networks, requires fine-tuning for different ML models.
[243]	Robust to Byzantine attacks and poisoning attacks	Adaptive filtering with a shadow dataset	Local Model Filter (LMF) with adaptive filtering policy to filter poisoned models based on performance on a shadow dataset	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	Scalability issues and computational overhead, especially when the ratio of malicious clients is high
[304]	Robust to Byzantine attacks	Supervising local models with a shadow dataset	Using a Local Model Filter (LMF) with accuracy and loss modes to filter out poisoned models	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	The approach requires a shadow dataset, and its effectiveness is influenced by the non-IID level of data.
[89]	Robust to model poisoning attacks and inference attacks	Blockchain-based FL	VPSA using Hyperledger Fabric to replace central curator and secure prediction mechanism	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental and Theoretical	High communication and computational overhead due to blockchain integration, scalability challenges, potential latency issues with larger networks.
[111]	Robust to model poisoning attacks, Byzantine attacks and inference attacks	Blockchain-based FL	Incorporates cross-validation to detect Byzantine attackers	Not specified	Custom	Experimental	High computational overhead, complexity in deployment due to the integration of multiple security mechanisms, potential scalability issues in large networks.
[226]	Robust to model poisoning attacks and Byzantine attacks	Decentralized Ring-Allreduce, Secret sharing	A decentralized deep FL scheme using Ring-Allreduce-based data sharing combined with threshold secret sharing	CV and HAR	MNIST, UCI-HAR	Experimental	High communication overhead due to ring structure, complexity in handling edge dropouts, scalability challenges in larger networks, especially with more complex datasets.
[299]	Robust to Byzantine attacks	Public auditing using a third-party auditor	FL-Auditor framework with sampling audits and lazy update mechanism	CV	MNIST, FEMNIST, CIFAR-10	Experimental	Potential overhead due to third-party auditing; scalability challenges in larger systems
[141]	Robust to Byzantine attacks	Blockchain-based hierarchical aggregation	VSS-based privacy-preserving robust gradient aggregation (VPRGA) integrated with blockchain to detect and eliminate anomalous gradients	CV	MNIST	Experimental	High communication and computational overhead due to secret sharing and blockchain integration, scalability concerns with larger networks
[103]	Robust to Byzantine attacks and model poisoning attacks	Reverse aggregation mechanism	Iterative elimination of compromised models based on validation loss decrease using a globally shared dataset	Text Classification	Google Speech Commands	Experimental	Potential limitations when handling highly non-IID data distributions; computational overhead in iterating the reverse aggregation process
[39]	Robust to Byzantine clients and poisoning attacks	Blockchain-based Stochastic model aggregation	Cre-DPSIGN algorithm integrating unbiased differential privacy mechanisms (CreFlip and CreGau) with blockchain for credit management and sampling	CV	MNIST	Both Experimental and Theoretical	Challenges in setting appropriate hyperparameters such as privacy budget and learning rate, affecting accuracy and robustness under different attack scenarios

Table 9. Manifold Hyperparameter Tuning during Model Training-based Treatments to Attain Robustness in FL (Part 1)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[313]	Robust to adversarial attacks	Client update mechanism	Maximizing the bias and variance during model update and learning the adversarially robust model updates during client update	CV	MNIST, Fashion-MNIST, CIFAR10, CIFAR100	Theoretical and Experimental	Limited non-IID cases; Rely on extra auxiliary adversarial datasets; Increasing communication cost
[234]	Robust to backdoor attacks	Introducing the ways of edge-case backdoor and how to resolve this issue	Detecting backdoor attack and the reason behind it to solve the issues by tuning the model in client-side	CV, OCR, Text prediction, Sentiment analysis	CIFAR10, ARDIS, and MNIST	Theoretical and Experimental	No backdoor-robust method proposed while giving an analysis of the hardness of detecting the edge-case backdoors and proposing a new backdoor attack
[23]	Robust to network errors	Hyper-dimensional learning-based training	To speed up training, reduce communication costs, and improve robustness to network faults, it conducts hyper-dimensional learning on features derived from a self-supervised contrastive learning framework instead of transmitting the model and simply training the hyper-dimensional component	CV	MNIST, Fashion-MNIST and CIFAR10	Experimental	No treatment on the models' aggregation; Limited experiments performed in the classical FL context; No experiments on non-IID data
[323]	Robust to real-world malicious data	Data sharing strategy and attention augmented model	The server pre-trains a global model with a public dataset. Clients in the FL framework obtain pre-trained model parameters and uniform data. Clients select their private data subset for FL training based on received model parameters and uniform data.	Travel mode identification from GPS trajectories	Geolife GPS trajectory	Experimental	Requires a uniform dataset in the server for training, Limited experiments performed in the classical FL context; No experiments on non-IID data
[40]	Robust to poisoning attacks	A defense mechanism leveraging adversarial learning	Employs produced samples to evaluate the activation of potential edge-case backdoors in the model and relies on generative adversarial networks to extract data characteristics from model changes	CV	MNIST, Fashion-MNIST, and CIFAR10	Experimental	Requires to generate augmented samples; Limited experiments performed in classical FL context; No experiments on non-IID data
[72]	Robust to sensor noise and environmental perturbations	Manual sanitary checks to verify the quality of data instances	Teachers use reliable instances as examples during teaching. The model, fine-tuned on these, aims for accurate predictions on them. Despite data corruption, the method integrates teaching and learning, producing a robust prediction model.	Regression	IJCNN, SUSY, CPUSMALL, and ABALONE	Experimental	Requires trusted instances; Limited experiments performed in classical FL context; Limited experiments on non-IID data
[311]	Robust to backdoor attack	Autonomously adjusts the learning rate	The method autonomously adjusts the learning rate based on the received parameter signs from all participants to reduce the impact of malicious updates	CV	FashionMNIST	Experimental	The hard threshold regulates the model accuracy; Limited experiments performed in classical FL context; No experiments on non-IID data
[159]	Robust to noisy federated weights	GAN-driven technique	A GAN detects fake federated weights, finding problematic local model updates before aggregation using min-max optimization	Content Caching	ML100K provided by MovieLens	Experimental	Limited experiments performed in classical FL context; No experiments on non-IID data, No experiment for channel noise variations and estimation for noise tolerance levels for federated weight averaging
[286]	Robust to Byzantine (poisoning) attacks	Extra logits-based predictive model	An adversarial setup adds a server-side predictive model for logit attribution. Simultaneously, the federated model is trained adversarially to counteract predictive behavior, reducing poisoning attack impact.	CV	MNIST, Fashion-MNIST, and CIFAR10	Experimental	Requires a different test dataset to generate the extra logits; No experiments on non-IID data; Unable to defend against collusion poisoning attacks and meanwhile protecting local gradients from leakage
[174]	Robust to data poisoning attacks	Model training using the existing FedAvg	Uses existing FedAvg approach to evaluate the impacts of the attacks	CV	GTSRB and Yale Faces	Experimental	Limited data poisoning attacks are considered; Limited non-IID settings
[279]	Robust to backdoor attacks	Model training using the existing approaches	Focuses on quantifying and understanding the implications brought by data heterogeneity in FL backdoor attacks	CV and sentiment classification	FEMNIST, Sent140, and CIFAR10	Experimental	Not considering non-IID settings; Limited considerations on aggregation strategies; No new development for robust FL
[255]	Robust to backdoor attacks with limited magnitude	Training using the existing approaches	Provides robustness guarantee from the theoretical aspects and conducts the related verification from the experimental aspects	CV and financial data classification	MNIST, EMNIST, and LOAN (financial datasets)	Theoretical and Experimental	No considering Non-IID settings; Limited considerations on aggregation strategies (only one) and other improved FL training paradigms

Table 10. Manifold Hyperparameter Tuning during Model Training-based Treatments to Attain Robustness in FL (Part 2)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[47]	Robust to model poisoning attacks and Byzantine attacks	Personalized training	Combines FL with personalized parameter learning strategy and multi-phase CT image analysis	Medical Imaging	Internal and external CT datasets including 1187 instances	Both Experimental and Theoretical	The study indicates potential biases in data collection and limitations due to retrospective study design. It also highlights the need for prospective validation.
[151]	Robust to adversarial attacks	Generative Adversarial Networks (GAN) for adversarial training	Combines GAN-based adversarial training with class probability distribution to balance natural accuracy and robustness	CV	MNIST, CIFAR-10, CIFAR-100, SVHN	Experimental	High computational cost, potential tradeoff between robustness and natural accuracy, requires tuning for different datasets
[260]	Robust to backdoor attacks	Graph Neural Networks (GNNs)-based training	Injection of subgraph triggers in Federated GNNs	Drug Discovery	NCI, PROTEINS, full, TRIANGLES	Experimental	Limited scalability to larger datasets and settings; Ineffectiveness of existing defenses in mitigating backdoor attacks
[288]	Robust to noisy labels	Noise-aware label correction and cross-validation sampling	Noise-aware Local Model Training Mechanism: Corrects noisy labels using a Label Correction Network (LCN) and meta-learning	CV	FashionMNIST, CIFAR-10, Clothing1M	Experimental	Increased computational overhead due to label correction, reliance on cross-validation that may not fully eliminate noise, scalability issues in large datasets
[85]	Robust to data poisoning attacks	Encrypted verification scheme	Verification key mechanism that encrypts and verifies data integrity before training	Not specified	Custom synthetic datasets	Experimental	Potential computational overhead, scalability challenges with large datasets, reliance on encryption that may introduce complexity
[128]	Robust to adversarial attacks	Adversarial training	Federated Adversarial Learning with Logits Calibration (FedALC) using square root inverse class frequency for local logits adjustment	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	High communication cost in adversarial training; Future work includes broader testing with more models and datasets
[306]	Robust to Byzantine attacks and untargeted poisoning attacks	Ensemble learning	Ensemble learning to train multiple global models and distance-based aggregation rule to mitigate outlier updates	CV	MNIST, Fashion-MNIST	Both Experimental and Theoretical	Current defenses struggle when a large proportion of clients are malicious; economic incentives for clients are often overlooked
[22]	Robust to adversarial attacks	Adversarial training	Uses adversarial examples in client training and aggregating local models	Spectrum sensing	Synthesized radar and wireless communication signals	Experimental	Needs further exploration in real-world settings for broader generalization
[102]	Robust to Byzantine/poisoning attacks	Contrastive learning and clustering	Ensemble of contrastive models trained with dimension reduction and clustering methods	CV	MNIST, CIFAR-10, FEMNIST	Experimental	Requires suitable computational power for training contrastive models, challenges in non-IID settings
[293]	Robust to adversarial attacks and Byzantine attacks	Adversarial training	Adversarial training to reduce adversarial success rate and transferability	CV	MNIST, CIFAR-10	Experimental	Balancing model utility and robustness; potential accuracy deterioration
[40]	Robust to backdoor attacks	Adversarial learning	ARMOR framework, including ARgan for generating synthetic datasets and MORpheus for attack mitigation	CV	MNIST, FashionMNIST, CIFAR-10	Experimental	High computational overhead, limited evaluation on diverse real-world scenarios, potential issues with GAN training stability
[81]	Robust to label noise	Spiral self-paced learning	Employs a spiral self-paced learning strategy to mitigate the impact of noisy labels	Medical Imaging	COVID Radiography dataset	Experimental	Potential computational complexity, challenges in scaling to larger and more diverse datasets, dependency on correct parameter tuning for self-paced learning
[54]	Robust to privacy inference attacks and poisoning attacks	Secure multiparty computation	SAFEFL framework integrating MPC with robust aggregation techniques	HAR	HAR	Experimental	High communication and runtime overhead; Limited testing on more complex architectures

Table 11. Client Selection-based Treatments to Attain Robustness in FL (Part 1)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[181]	Robust to stragglers and adversaries	Staleness-aware grouping with entropy-based filtering and loss-weighted averaging	Utilizes outcomes sent by slower components by initially categorizing models from the same original models (with identical delay), acquiring a representative model per group. Then, aggregates these representative models, considering their delays, to derive the global model.	CV	MNIST, Fashion-MNIST, and CIFAR10	Theoretical and Experimental	The server should have access to every received individual model to calculate the entropy and loss values
[238]	Robust to clients' malicious data	Client selection followed by the modification of models averaging	Model filtering for potential client selection and use of improved multi-Krum with homomorphic encryption for model averaging	Internet of Vehicle using CV dataset	MNIST	Theoretical and Experimental	Not considering other attack-robust aggregation baselines; Very limited Non-IID settings; Limited FL hyperparameters tested; Limited attack cases; Just a simple combination for existing aggregation method
[147]	Robust to corrupted models	Client selection	Clients' contributions gauge via layer-wise network parameter distance to the server. Standard-range distances carry higher weight. Short distances imply offline status, affecting global model convergence.	CV	FEMNIST	Experimental	High computation cost on the server; Not considering non-IID case; Relatively poor performance on non-CNN models; Limited FL hyperparameters tested; Limited corrupt model cases
[88]	Robust to adversarial models	A new consensus mechanism named PoLS to detect adversaries	Model interpretation, contribution calculation, and label-wise aggregation for effective client selection	CV	MNIST, Fashion-MNIST, and CIFAR10	Theoretical and Experimental	Just considering data poisoning attacks without model poisoning attacks
[119]	Robust to Byzantine-attacks	Client selection	Enhances clustering-based scheme robustness via automatic update clipping. Clipping occurs before clustering using clip by norm, adjusting vectors that surpass a preset value.	CV	MNIST, CIFAR10	Experimental	Poor performance on non-IID data; Limited improvement against Byzantine-attacks
[140]	Robust to faulty and noisy clients	Client selection	Addresses information leakage, lessons dilution from normal clients' updates, and factors in whole population distribution during client selection	CV	MNIST	Theoretical and Experimental	Proposed a backdoor attack instead of a robust FL
[201]	Robust to model poisoning backdoor attack	Robust filtering of one-dimensional outliers	Filtering out the malicious clients' model updates via a univariate outlier detection technique	CV	FEMNIST, CelebA-S, and CelebA-A	Experimental	Limited experiments performed in the classical FL context; No experiments on non-IID data
[67]	Robust to Byzantine-attacks	Filter useful updates, excluding malicious clients	Benign nodes are chosen based on Euclidean distance from neighbors, utilizing their own baseline rather than neighbor mean/median for calculation	CV	MNIST, and CIFAR10	Theoretical and Experimental	It is good in low computational overhead in any system, but in high computational overhead, it works normally. No experiments on non-IID data
[259]	Robust to Byzantine model poisoning attacks	Sign-gradient filtering to identify malicious gradients	It uses two gradient filters (norm-based filtering and sign-based clustering) to select the trusted gradient set; then aggregate the global gradient with existing aggregation rules, like trimmed-mean, and so on	Image and Text Classification	MNIST/Fashion-MNIST, CIFAR10, and AG-News	Theoretical and Experimental	Limited Non-IID setting (strictly its so-called Non-IID setting in the paper is a variation of IID); poor performance in the Non-IID settings
[239]	Robust to data poisoning attacks	Data-complexity-based clients' batch detection method	To counter FL data poisoning, a 3-stage method is introduced: (1) Utilize data complexity features to form a training dataset for binary classification. (2) Train a layered detection model. (3) Detect poison data locally.	Spam Detection	UCI's Spam dataset	Experimental	No considering non-IID settings; Unrealistic training of poison data batch detection model; Just consider one data poison case (label flipping); Just one dataset consideration on horizontal FL
[130]	Robust to malicious client	Gradient-based binary permutation client selection	Uses gradient-based binary permutation to create super nodes within a base station. Implements training and group aggregation via one-step internal synchronization. Achieves group model aggregation via multistep external synchronization	CV	FEMNIST	Theoretical and Experimental	Just one dataset considered; Limited non-IID settings; Potential failure as facing highly skewed non-IID settings within BSs; Particular FL applications
[227]	Robust to data poisoning attacks	PCA-based malicious detection method	Calculates gradient variation vs. initial global model for each local model. Applies PCA for 2D reduction. Identifies malicious gradients in reduced dimensions, excluding them from global aggregation.	CV	CIFAR10, and FMNIST	Experimental	No non-IID, only one attack considered; Limited defense due to linear PCA's difficulty in capturing representative low-dimensional local gradient features
[145]	Robust to local model poisoning attack	Client selection	This includes 4 stages: (1) Pre-aggregation: use attack-specific aggregation for local models; (2) Detection: employ Pearson correlation for Krum, and parameter removal count for trimmed-mean and median; (3) Removal: discard malicious models; (4) Re-aggregation: aggregate rest models.	CV	FEMNIST	Experimental	Limited model poisoning attacks are considered; No non-IID consideration; Just one dataset tested; Poor generalization for the proposed defense method
[202]	Robust to poisoning attacks	Client selection	This method employs the Induced Ordered Weighted Averaging operator for client aggregation. Weights are determined by an induced-ordered function and linguistic quantifier, considering local model effectiveness and predefined factors.	CV	MNIST, FMNIST, and CIFAR10	Experimental	Require access to the raw data to design the client selection mechanism; Limited experiments performed in the classical FL context; No experiments on non-IID data

Table 12. Client Selection-based Treatments to Attain Robustness in FL (Part 2)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[302]	Robust to model replacement attacks and backdoor attacks	Reverse engineering and global reverse trigger-based malicious clients detection	The method consists of four steps: model correction, outlier identification, global reverse trigger creation, and reverse engineering. Clients in FL reverse engineer the global models they get before each training session.	CV	MNIST, and GTSRB	Experimental	Increases computation and communication costs; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[70]	Robust to failure of edge devices	Client selection	Data volume, computation power, and network bandwidth have been used as the parameters to compute the optimal client set	CV	MNIST	Experimental	Requires access to the raw data to design the client selection mechanism; Limited experiments performed in the classical FL context; No experiments on non-IID data
[222]	Robust to poisoning attacks	GAN-based countermeasure algorithms to avoid anomaly in aggregation	The approach calculates anomalous values per client and excludes those above the threshold from aggregation	IIoT	Drebin, Contagio, and Gnome	Experimental	Requires access to the raw data to design the GAN-based clients detection; Limited experiments performed in the classical FL context; No experiments on non-IID data
[295]	Robust to model poisoning attacks	Detecting malicious clients via checking their model-updates consistency	Based on previous model updates, the server forecasts a client's model update for each iteration and flags a client as malicious if the received model update and the projected model update differ repeatedly	CV	MNIST, CIFAR10, and FEMNIST	Experimental	No discussion on the expansion of vertical FL; Asynchronous FL in different domains including text and graphs, and effective recovery of the global model from model poisoning assaults after detecting the identified malicious clients; Limited experiments on non-IID data
[17]	Robust to malicious clients	Trust Bootstrapping-based malicious clients detection	The server uses a clean root dataset to build a base model. Clients' updates receive trust scores based on alignment with the server's update direction. Normalizing local update magnitudes averaging updates are weighted by trust scores for global model updates.	CV and Human Activity Recognition	MNIST, CIFAR10, and Human Activity Recognition	Theoretical and Experimental	Require a clean dataset in the server; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[69]	Robust to malicious and unreliable clients	History of gradients-based malicious and unreliable clients detection	The key idea is to employ long-short histories of gradients along with carefully selected distance and similarity metrics during the iterative model update process	CV	MNIST, and Fashion-MNIST	Experimental	No consideration of backdoor attacks; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[157]	Robust to membership and reverse attacks	Filtering out anomaly parameters from Byzantine adversaries	To provide privacy and filter out abnormal parameters from Byzantine attackers, the proposed method is built from an existing Byzantine-robust FL method and supplemented with distributed Paillier encryption and zero-knowledge proof	CV	MNIST, EMNIST, and Fashion-MNIST	Experimental	Limited experiments performed in the classical FL context; No experiments on non-IID data
[163]	Robust to poisoning attacks	Similarity measure to detect malicious gradients of the clients	Cosine similarity is used to judge the malicious gradients uploaded by malicious clients	CV	MNIST, and Fashion-MNIST	Theoretical and Experimental	Limited experiments performed in the classical FL context; No experiments on non-IID data
[52]	Robust to imperfect channel state information	Device selection, transmit power allocation, and receive beamforming	In the presence of channel uncertainty, the transmit power and receive beamforming are determined by minimizing the maximum MSE. Devices with minor errors are chosen until the overall MSE is near to, but not greater than, the specified number, the acceptable maximum aggregate error	CV	MNIST	Theoretical and Experimental	The hard threshold for tolerable maximum aggregation regulates the model accuracy; Limited experiments performed in the classical FL context; No experiments on non-IID data
[7]	Robust to noise in the models during communication steps	Sampling-based successive convex approximation	A feasible training scheme is created using a sampling-based successive convex approximation approach due to the non-convex challenges of the objective function and the unavailable maximum noise condition	CV	MNIST	Theoretical and Experimental	No consideration of non-IID data issue in the theoretical proof; Very limited experiments performed in classical FL context; No experiments on non-IID data
[300]	Robust to Byzantine (poisoning) attacks	Model-level defensive mechanism based on poisoned model detection	It selects vital model parameters, offering a concise representation of local model inputs. An online unsupervised detection approach identifies and excludes poisoned models from joining the global intrusion detection model.	Network intrusion detection	UNSW-NB15, CICIDS2018	Experimental	Limited poisoning attacks tested; Limited experiments performed in classical FL context; Limited experiments on non-IID data
[257]	Robust to Byzantine (poisoning) attacks	Robust truth discovery aggregation scheme to remove malicious model updates	Assigns weights according to users' contribution and uses maximum clique-based filter aggregation to guarantee global model quality	CV	MNIST, Fashion-MNIST, CIFAR10, and Clothing1M	Theoretical and Experimental	Limited experiments on non-IID data; Unable to maintain high accuracy for Byzantine majority settings; No incentive mechanism that encourages more users to participate honestly-majority settings
[144]	Robust to label-flipping attacks	Cloud-side malicious node detection mechanism	Detects malicious nodes by testing the local model quality using a specific testing dataset to test the sub-models uploaded by edge nodes to detect malicious nodes	CV	MNIST, and CIFAR10	Experimental	Requires a local test dataset in each client; Limited poisoning attacks tested; Non-IID data setting is not considered
[308]	Robust to Byzantine attacks	Malicious client detection mechanism	Identifies malicious clients by embedding a watermark to the global model and tracking the degree of watermark recession after local model training	CV	MNIST, and CIFAR10	Experimental	The detection mechanism is not optimized; Limited (two) attacks tested, Non-IID dataset is not considered

Table 13. Client Selection-based Treatments to Attain Robustness in FL (Part 3)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[104]	Robust to model poisoning attacks and malicious and unreliable clients	History-based anomaly detection	Mitigates on-off, bad-mouthing, and good-mouthing attacks through adaptive trust scaling and anomaly detection	CV and Intrusion Detection	MNIST, CIFAR-10, KDD Cup '99	Both Experimental and Theoretical	Potential overreliance on history-based mechanisms that might be vulnerable to advanced evasion techniques; scalability concerns in extremely large networks
[34]	Robust to Byzantine failures and model poisoning attacks	Client selection	Presents a dynamic gradient filter algorithm to filter out malicious updates	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	The effectiveness of DGF could be reduced in high-dimensional spaces; potential reduction in efficiency in targeted attacks
[42]	Robust to label-flipping attacks	Robust aggregation and client selection	Adjusting local updates and benchmark updates, using preference values for aggregation, enhancing client selection based on loss reduction and historical performance	CV and Text Classification	Fashion-MNIST, Agnews	Experimental	The need for further adjustments for NLP tasks to enhance robustness; performance could vary based on dataset characteristics and parameter settings
[225]	Robust to model poisoning attacks and label-flipping attacks	Cosine similarity-based detection	Combines personalized logit adjustment for non-IID data and server-side poisoned client detector using cosine similarity	Intrusion detection	N-BaIoT (IoT traffic data)	Experimental	Potential computational overhead, effectiveness may diminish with high proportions of poisoned clients (>30%), challenges in extreme non-IID settings
[134]	Robust to Byzantine attacks	Client selection	Cosine similarity-based Byzantine detection	CV	MNIST, CIFAR-10	Both Experimental and Theoretical	High computational complexity due to encryption; Challenges in scalability with large datasets and client numbers
[179]	Robust to model poisoning attacks	Hierarchical aggregation with reputation-based client selection	Selecting clients for aggregation based on reputation scores	CV and Medical Imaging	MNIST, Fashion-MNIST, Chest Xray	Experimental	Needs exploration in real-time settings with dynamic data and varying client availability
[266]	Robust to model poisoning attacks	Client selection	Introduces a new metric called GradScore to detect malicious clients by measuring the gradient norms	CV	MNIST, CIFAR-10	Experimental	Relies on convergence to detect malicious activity; the impact on other types of attacks like label-flipping not discussed
[95]	Robust to malicious clients	Client selection	Novel dynamic reputation protocol to detect and eliminate adaptive participants	CV	MNIST, MedMNIST	Experimental	Did not address potential alternating behavior of participants; Vulnerability to 51% attack
[68]	Robust to Byzantine attacks and poisoning attacks	Client selection	Use of element-wise sign function on local model updates, Poisoning Attack Detector to filter out malicious updates	CV	MNIST, F-MNIST, CIFAR-10	Experimental	Limited to specific types of poisoning and privacy attacks, may not generalize well to other adversarial settings, potential tradeoff between model accuracy and robustness due to masking
[63]	Robust to malicious and unreliable clients, and model poisoning attacks	Anomaly detection using autoencoders, Incentive mechanism based on credit scores	Fourier transform-based anomaly detection, Credit score-based client incentive mechanism	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	High computational complexity for anomaly detection, potential issues with scalability in large networks, need for real-time detection mechanisms.
[154]	Robust to model poisoning attacks, and malicious and unreliable clients	Client selection	Groups clients using hypermesh, employs secret sharing and Pederson commitment for detecting and removing malicious clients	CV	MNIST, CIFAR-10, CelebA	Experimental	Potential computational overhead, scalability challenges in large networks, complexity in implementing secret sharing and commitment mechanisms
[242]	Robust to backdoor attacks	Client selection	SCFL framework using SVD for feature extraction and k-means clustering to filter out malicious updates	CV and Text Classification	MNIST, CIFAR-10, Amazon Fine Food Reviews	Experimental	Potential issues with scalability and complexity; Sensitivity to clustering parameters and Non-IID settings
[57]	Robust to poisoning attacks and Byzantine attacks	Amplification of gradients	Two approaches: AGRMP (Max Pooling) and AGRXAI (Explainable AI) to amplify and detect malicious updates	CV and Text Classification	CIFAR-10, MNIST, Fashion-MNIST, CATvsDOG, PURCHASE100, LOCATIONS30, TEXAS100	Experimental	The need to balance robustness, fidelity, and efficiency; further work required to reduce the time complexity for AGRXAI
[146]	Robust to data poisoning attacks, model poisoning attacks and Byzantine attacks	Client selection	Use of Wasserstein distance to identify outlying data commitments and asynchronous model convergence checks	Not specified	Two non-IID datasets	Experimental	Limited by effectiveness in highly non-IID data settings, potential fabrication of commitments by attackers
[167]	Robust to model poisoning attacks and malicious and unreliable clients	Client selection	PCA for dimensionality reduction, Binary Tree-based Clustering with Noise (BTBCN), Self-Ensemble Detection Correction (SEDC)	CV	MNIST, EMNIST, CIFAR10, COVIDx	Experimental (extensive)	Requires setting parameters like min cluster size, potentially limited scalability to larger and more complex models

Table 14. Client Selection-based Treatments to Attain Robustness in FL (Part 4)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[138]	Robust to model poisoning attacks (both targeted and untargeted)	Detection and punishment of malicious clients	Detects malicious clients using cosine distances of local updates and punishes them based on historical behavior	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	Challenges in defending against adaptive attacks; requires fine-tuning for non-IID data settings
[183]	Robust to model poisoning attacks	Client selection	Uses Shapley values for smart client selection	CV	FEMNIST, CelebA, Sent140, CIFAR-10	Experimental	Complexity in computing Shapley values, potential scalability issues, challenges in handling extreme cases of data heterogeneity
[36]	Robust to Byzantine attacks and model poisoning attacks	Client selection	A Byzantine-robust FL scheme leveraging 3PC to securely select and aggregate models while preserving privacy	CV and Medical Imaging	CIFAR-10, MNIST, PathMNIST, Lung Cancer	Experimental and Theoretical	High computational overhead, potential scalability issues with large-scale datasets, complexity in implementation due to cryptographic techniques
[173]	Robust to Byzantine attacks and model poisoning attacks	Client selection	Dual filter mechanism to identify and discard malicious gradients, adaptive weight adjustment to reduce the impact of suspicious gradients, and dynamic clipping to scale gradients	CV	MNIST, Fashion-MNIST, CIFAR-10, Tiny-ImageNet	Experimental	Increased computational complexity due to the filtering and clipping mechanisms, scalability challenges in large IoT networks, dependency on the correct setting of threshold parameters
[30]	Robust to unreliable clients	Gradient Recycling	FL with Gradient Recycling (FL-GR)	CV	MNIST, CIFAR-10, CIFAR-100	Both Experimental and Theoretical	Assumption of reliable downlink, potential high memory usage
[132]	Robust to poisoning attacks	Voting and scaling of local updates	Voting mechanism for filtering abnormal directions; Scaling mechanism for adjusting abnormal magnitudes	CV and HAR	MNIST, Fashion-MNIST, CIFAR-10, BCW, HAR	Experimental	Difficulty in converging the global model under high non-IID settings; Potential vulnerability to adaptive attacks
[284]	Robust to Byzantine attacks and stragglers	Client selection	Uses cosine-similarity-based Byzantine-robust aggregation	CV	MNIST, CIFAR-10	Experimental	High recovery threshold; Computational overhead for Lagrange coded computing
[271]	Robust to Byzantine attacks and poisoning attacks	Enclave-aided robust aggregation	Evaluating layer-wise cosine similarity between local and guide models, using boxplot for outlier detection	CV	MNIST, CIFAR-10	Experimental	Performance may degrade under high attack scenarios; efficiency improvements needed for larger models when using TEE
[14]	Robust to Byzantine attacks and label skewness	Two-stage robust aggregation	Stage 1 involves finding a robust subspace; Stage 2 estimates the simplex and filters outliers	CV and Text Classification	MNIST, CIFAR-10, AG-News	Experimental	Needs further exploration in more complex non-IID settings and larger datasets; potential scalability issues under extreme label skewness scenarios
[33]	Robust to model poisoning attacks and Byzantine attacks	Robust and Dynamic Gradient Filtering	The RDGF algorithm screens out malicious updates by calculating distance-based scores for client gradients, and uses geometric median aggregation and noise addition to enhance robustness	CV	MNIST	Experimental	The effectiveness of RDGF may decrease in scenarios with high variability in client behavior or extreme levels of data heterogeneity. Further optimization for varying attack patterns is needed.
[169]	Robust to Byzantine attacks	Saliency Maps and Structural Similarity Index (SSI) for detecting gradient anomalies	Compares Saliency Maps using SSI to identify and isolate contributions from potentially malicious clients	CV	CIFAR-10, MNIST, Fashion-MNIST	Experimental	Potential computational overhead due to the calculation of Saliency Maps and SSI, challenges in scaling to larger datasets or models
[312]	Robust to evasion attacks and poisoning attacks	Feature center separation, Outlier detection	Incorporates FeaS to increase feature margins between classes and Malicious Center Detection (MaID) using Gaussian Mixture Models to detect and remove malicious clients	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	Potential computational overhead due to feature separation and outlier detection mechanisms, scalability challenges in large datasets, dependency on correct configuration of the detection mechanism
[264]	Robust to data and model poisoning attacks	Clustering-based aggregation, Cosine similarity	Robust Federated Clustering with cosine similarity to select similar cluster models and defend against poisoning attacks	CV	MNIST, CIFAR-10, Fashion-MNIST	Experimental	High complexity in clustering and cosine similarity calculations, potential scalability challenges, sensitivity to the number of attackers and data heterogeneity
[100]	Robust to model poisoning attacks and backdoor attacks	Client selection	DeMAC system: detecting malicious clients by measuring GradScore	CV	MNIST, CIFAR-10	Experimental	Detection might be less effective under extremely low poisoned data rates or highly non-IID settings
[272]	Robust to model poisoning attacks, Byzantine attacks and label-flipping attacks	Adaptive model filtering based	Dynamic selection of cosine similarity and Euclidean distance-based filtering mechanisms with weighted aggregation	CV	MNIST, CIFAR-10	Experimental	Potential performance degradation in high attack scenarios with non-IID data; needs further optimization to balance robustness and efficiency
[88]	Robust to model poisoning attacks and Byzantine attacks	Blockchain-based reputation-aware client selection	PolS consensus mechanism based on client contributions, reputation, and past performance	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	The need to address communication overhead and scalability in large-scale implementations; further research needed on integrating data security with model interpretation

Table 15. Client Selection-based Treatments to Attain Robustness in FL (Part 5)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[55]	Robust to Byzantine attacks and poisoning attacks	Synthesized trust with anomaly detection	Trust synthesizing mechanism combining trust scores (TS) and confidence scores (CS) with dynamic trust ratio	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	Potential issues with the root dataset's bias; need for further optimization of the trust synthesizing mechanism
[58]	Robust to Byzantine/poisoning attacks	Client selection	Collaborative blind cross-evaluation mechanism and smart activation mechanism for detecting and isolating malicious nodes	Medical Imaging	ECG dataset	Both Experimental and Theoretical	The approach may lose iterations during cross-checking, leading to slower convergence; potential risk of false alarms during convergence; challenges in distinguishing between legitimate and Byzantine nodes
[298]	Robust to Byzantine attacks and label-flipping attacks	Client selection	Intermediate parameter compression, Update quality evaluation mechanism using cosine similarity between model updates	CV	MNIST	Both Experimental and Theoretical	The approach might lead to slower convergence and potential false alarms during model aggregation; scalability could be a challenge with an increased number of nodes
[249]	Robust to model poisoning attacks and malicious and unreliable clients	Client selection	Uses gradient similarity for malicious detection, contract theory for fair incentive distribution among participants	CV and Text Classification	MNIST, CIFAR-10, Movie Review dataset (MR)	Experimental	Computational complexity in gradient similarity calculations, potential challenges in scalability and real-time detection, balancing fairness and security in diverse networks
[27]	Robust to data and model poisoning attacks	Clustering, cosine similarity, meta-learning	RFCL (Robust Federated Clustering Learning): Groups clients' models into clusters based on cosine similarity and uses meta-learning to aggregate models from the most similar clusters	CV	MNIST, CIFAR-10, Fashion-MNIST	Experimental	Computational complexity due to clustering, potential challenges in handling large-scale heterogeneous data, reliance on accurate clustering and similarity measures
[232]	Robust to Byzantine attacks	A four-pronged defense mechanism to mitigate Byzantine attacks	Reliable client selection using absolute similarity and spectral-based outlier detection	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	1. Suboptimal performance when attackers dominate, 2. Lack of theoretical analysis to support the findings
[25]	Robust to Byzantine attacks, poisoning attacks and label-flipping attacks	Attention-based aggregation	Combines scaled dot-product attention mechanism with vector filter to weigh and filter gradients	CV	MNIST, FMNIST, CIFAR-10	Experimental	Needs further refinement to handle scenarios with more than 50% malicious clients, and potential improvements in efficiency and scalability under heavy attack conditions
[292]	Robust to Byzantine attacks and poisoning attacks	Resilience through AGR	Optimizes adversarial impact while evading detection by AGRs	CV and Text Classification	CIFAR-10, Purchase100, Texas100, Location30, 1000 Genome Project	Experimental	Further research needed to develop defense mechanisms specifically targeting PMIA in various FL settings and to explore the implications of AgrEvader under different threat models
[229]	Robust to poisoning attacks and Byzantine attacks	Blockchain-based decentralized aggregation	Use of a decentralized blockchain committee for model aggregation	CV	MNIST, CIFAR-10	Experimental	Complexity of the validation process; potential vulnerabilities in the exposure of next round's committee members leading to targeted attacks on aggregators
[235]	Robust to model poisoning attacks and Byzantine attacks	Dynamic Clustering-based Aggregation	Sign Consistency Ratio (SCR) and Norm-based adaptive clipping to filter out malicious updates	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	The method may be less effective under extreme non-IID conditions, and there might be an increased computational cost due to clustering and adaptive clipping processes
[61]	Robust to Byzantine attacks and backdoor attacks	Anomaly detection	VAE models for detecting and excluding malicious model updates (spatial perspective), and a momentum-based update filter to generate robust global model updates	CV	MNIST, MNIST-D, FEMNIST	Experimental	Potential limitations in handling highly variable fractions of malicious clients and challenges in generalizing across different datasets and tasks
[166]	Robust to model poisoning attacks and malicious and unreliable clients	Anomaly detection	Integrates anomaly detection to filter out malicious updates	CV	MNIST, CIFAR-10	Both Experimental and Theoretical	Potential high communication and computation overhead; scalability concerns
[301]	Robust to model poisoning attacks, Byzantine attacks and label-flipping attacks	Anomaly detection and aggregation filtering	Use of a double-aggregation mechanism combined with weighted averaging to filter out anomalies	CV	MNIST, CIFAR-10, Fashion-MNIST	Experimental	Potential high computation costs due to double aggregation; may struggle with highly sophisticated attacks
[296]	Robust to backdoor attacks	Client selection	Estimates data divergence among clients to detect backdoor clients	CV and Text Classification	SST-2, IMDB, Amazon, AgNews	Experimental	The defense performance may degrade under non-IID settings and when adaptive attacks are specifically designed against the proposed method
[319]	Robust to noisy and unreliable clients	Adaptive client selection	A Bayesian framework is used to estimate client utility by integrating the weight parameters of local models and their performance	CV and Speech Classification	CIFAR10, Google Speech, HARBox	Experimental	The effectiveness of the approach in other FL settings or with other types of data noise is not explored

Table 16. Client Selection-based Treatments to Attain Robustness in FL (Part 6)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[248]	Robust to noisy label learning, class imbalance, and label noise heterogeneity	Noisy client detection	Per-class loss calculation for noisy client detection using Gaussian Mixture Model	Medical Imaging	ICH (Intracranial Hemorrhage) dataset, ISIC 2019 dataset	Experimental	Not specified explicitly; may require further investigation into different data heterogeneity patterns beyond the medical domain
[240]	Robust to Byzantine/poisoning attacks	Isolation Forest for anomaly detection	Detects malicious model updates using Isolation Forest after dimensionality reduction	CV	MNIST, CIFAR-10, Fashion-MNIST	Experimental	Potential over-reliance on dimensionality reduction, impact on convergence not fully explored
[91]	Robust to data poisoning attacks and poisoning attacks	Hyperdimensional Computing (HDC) for encoding updates	Encodes local model updates using HDC, uses cosine similarity-based aggregation to detect and filter out poisoned updates	CV	MNIST, CIFAR-10	Experimental	Potential computational overhead in encoding/decoding, challenges in non-IID data settings, possible degradation in non-IID accuracy
[115]	Robust to model poisoning attacks, Sybil attacks, and untargeted and targeted attacks	Client selection	Private evaluation of local models, hierarchical clustering for outlier removal	CV and Text Classification	FMNIST, CIFAR, TREC, AGNEWS	Experimental	High computational and cryptographic overhead, complexity in implementing and verifying zk-SNARKs, challenges in scaling to larger datasets and more complex models
[196]	Robust to backdoor attacks and model poisoning attacks	Graph-theoretic approach using correlation analysis	Identifies and isolates backdoor attackers by analyzing the correlation between clients' weight updates	CV	Fashion-MNIST	Experimental	Assumes sufficient correlation difference between benign and malicious clients, may face challenges with highly sophisticated adversaries, and potential computational overhead
[4]	Robust to data and model poisoning attacks	Clustering with cosine similarity, Meta-learning	Uses clustering to group client models, cosine similarity to select high-quality clusters, and meta-learning for personalization and aggregation	CV	MNIST, CIFAR-10, Fashion-MNIST	Experimental	High computational complexity due to clustering and similarity analysis, potential scalability challenges with larger datasets or more diverse networks, dependency on appropriate parameter tuning
[251]	Robust to malicious client attacks	Dynamic feature extraction and clustering	Sparse representation-based feature extraction and optimized k-means clustering for identifying and excluding malicious clients	Medical Imaging	Organ-MNIST Axial	Experimental	Limited resistance to data poisoning under severe label skew conditions
[244]	Robust to malicious attacks	Detection and blocking of malicious updates	RoFL scheme using counters, validators, and authenticators to identify and exclude malicious clients	CV	MNIST	Experimental	The detection rates decrease as the number of malicious clients increases; limited validation on more complex datasets
[210]	Robust to model poisoning attacks and Byzantine attacks	Siamese networks and anomaly detection	A Siamese network-based anomaly detection framework using CNNs to identify malicious updates	CV	FEMNIST dataset	Experimental	Potential computational complexity and scalability issues when using CNNs for model detection

Table 17. Modification of Model Aggregation-based Treatments to Attain Robustness in FL (Part 1)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[228]	Robust to Byzantine clients	Modified arithmetic averaging by client's weight adjustment	Detecting the probability distribution of the softmax layer and then penalizing the client models' weights with abnormal distribution for aggregation	-	-	-	Limited FL hyperparameters combinations tested; Limited Non-IID cases considered; Poor performance on the highly skewed non-IID case
[117]	Robust to malicious clients	Geometric-median-based aggregation	Geometric-median-based aggregation is performed after re-weighting the models via excluding extreme outliers based on a user-defined skewness threshold	IoT	FEMNIST and Bosch IIoT	Theoretical and Experimental	Limited data and model poisoning attacks tested; Limited FL hyperparameters combinations tested; Limited Non-IID cases considered
[274]	Robust to Byzantine failure	Two techniques: median-based GD and trimmed-mean-based GD	To make distributed learning more robust against Byzantine failures, the server has two options to do the global gradient update: (i) utilize the median-based gradient descent method; or (ii) use the trimmed-mean-based gradient descent method	CV	MNIST	Theoretical and Experimental	Not considering Non-IID setting; Simple experimental setting without comprehensive parameter configurations and analysis; Only one dataset tested; Limited technical innovation as two main GD methods are not originally proposed by the authors; Limited Byzantine failures considered
[162]	Robust to corrupted models	Geometric-median-based aggregation	Trimmed softmax is used as the weighting function. In essence, the aggregation yields a global prediction that minimizes the total of the weighted Euclidean distances between the global forecast and each specific local prediction	CV	FEMNIST	Experimental	Wrong exclusion could lead to significant damage to the model's performance; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[15]	Robust to outlier models	Key-Resolver using medians-based aggregation	Krum approach identifies a single key model that is considered the most reliable and accurate among the group. This key model is then used to make the final decision. The Krum approach aims to mitigate the negative effects of potentially faulty or malicious models by selecting a key model that is less likely to be compromised.	CV	MNIST	Theoretical and Experimental	Sensitive to outliers and models with distinct outputs, impacting key model selection; The chosen distance metric greatly influences Krum approach performance; Limited experiments were performed in the classical FL context; No experiments on non-IID data
[135]	Robust to Byzantine attackers	Trimmed padding mean-based aggregation	The server calculates the median of uploaded parameters and uses it as the optimal global model. For the majority of benign clients, this median aligns with good client values. The server replaces trimmed parameters, maintains trimmed mean with a fixed ratio, and aggregates using averaging for the final global model.	CV	MNIST	Experimental	No study how to maintain privacy and robustness simultaneously in the case of non-IID scenario; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[187]	Robust to outlier updates	Use of the geometric median	The method is based on a robust aggregation oracle based on the geometric median, which produces a robust aggregate by iterating a normal non-robust averaging oracle a certain number of times	CV and language modeling	MNIST, CIFAR10, and Shakespeare	Theoretical and Experimental	There is no investigation of resilience in the presence of heterogeneity and personalization
[231]	Robust to outlier updates	Use of the node-wise trimmed mean	Pruning a small percentage of extreme (lowest and highest) weights at each node of the local models to perform the node-wise trimmed mean	CV and language modeling	MNIST, CIFAR10, and Shakespeare	Theoretical and Experimental	Direct pruning of weights may impede the intended robust aggregation
[206]	Robust to label-flipping attack	Zeno and coordinate-wise median aggregation	Zeno ranks clients based on loss function descent predictions for pRef. Higher-ranked clients are aggregated. Coordinate-wise median resembles average aggregation but combines weights by local model weight medians.	Device identification	RPi model device identification	Experimental	Classical adversarial attacks (evasion attacks or model poisoning attacks) not tested; Limited experiments performed in the classical FL context; No experiments on non-IID data
[2]	Robust to the variations of estimate in models parameters	Hessian matrix-based aggregation mechanism	The Hessian as a scaling matrix is used in a way analogous to Quasi-Newton approaches by utilizing the first-order information of the loss function. The influence of different levels of data quality across local models is captured by this treatment.	CV	MNIST, Fashion-MNIST, and CIFAR10	Experimental	Increases communication cost as the Hessian matrix of the models need to be communicated among the server and clients; Limited experiments on non-IID data
[223]	Robust to adversarial attacks	Aggregation algorithm inspired by the truth inference methods	A truth inference framework estimates client update reliability to remove outliers before aggregation, countering data poisoning. Multi-round historical parameters enhance robust aggregation.	CV	MNIST, Fashion-MNIST, and CIFAR10	Experimental	Limited poisoning attacks tested; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[160]	Robust to targeted and untargeted model poisoning attacks	Weights divergence calculation-based aggregation	Sanitizing factors based on temporal divergence and historical behaviors compute participant divergence, used to sanitize local model parameters during aggregation in the proposed method	CV	MNIST, and CIFAR10	Experimental	Collusive poisoning attacks become unbeatable when the proportion of adversarial participants is relatively large, like about 50%; No experiments on non-IID data
[303]	Robust to Byzantine (poisoning) attacks	Median-based aggregation followed by Euclidean distance-based client sampling	Introduces a client sampling scheme for estimating adversary probability based on Euclidean distances among sampled vectors. It proposes using coordinate-wise median from honest clients to eliminate malicious updates effectively.	CV	MNIST, and CIFAR10	Theoretical and Experimental	Experiments conducted using Intel SGX's architecture in a LAN environment has limited memory capacity; Very limited experiments on non-IID data
[203]	Robust to backdoor model poisoning attacks	Aggregation rule modification	Analyzes several Byzantine-Tolerant aggregation methods including Krum, Multi-Krum, RFA, and Norm-Difference Clipping	CV	CIFAR-10, and EMNIST	Experimental	Not considering non-IID settings; Limited considerations on aggregation strategies; No new development for robust FL

Table 18. Modification of Model Aggregation-based Treatments to Attain Robustness in FL (Part 2)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[320]	Robust to Byzantine attacks and model poisoning attacks	Cosine Similarity Aggregation	Convolutional Kernel Angle-based Defense Aggregation	CV	CIFAR-10, Fashion-MNIST	Experimental	Challenges in distinguishing between malicious and normal gradients in high-dimensional space; may require substantial computational resources for large models
[168]	Robust to model poisoning attacks	Randomization and aggregation techniques	Switch, Layered-Switch, and Weighted FedAvg aggregation functions	CV	CIFAR10, CIFAR100, Fashion-MNIST	Experimental	Limited testing on other datasets, scalability of the approach
[262]	Robust to Byzantine-resilient, Data poisoning attacks, Backdoor attacks	Integration of robust statistical methods	Integrates Robust Normal Aggregation (RN) module, Personalized Fusion (PF) module, normality tests, probabilistic perturbations	CV	MNIST, CIFAR-10	Experimental	Further refinement of defense capabilities against more complex Byzantine and poisoning attacks is needed
[32]	Robust to model poisoning attacks, Byzantine attacks, privacy attacks	Quadratic Voting (QV) mechanism for aggregation	Aggregates updates based on QV, penalizes abnormal similarity scores, and allocates unequal budgets to parties based on reputation	CV	MNIST, Fashion-MNIST, FEMNIST, CIFAR10.	Both Experimental and Theoretical	Limited to specific types of attacks (model and data poisoning); may require integration with other Byzantine-robust techniques for broader robustness; computational overhead due to QV and reputation-based budgeting
[291]	Robust to Byzantine attacks	M-estimation with robust aggregation rules (e.g., Median, Trimmed Mean, Krum)	Combines gradient tracking and proximal algorithm with robust aggregation	Text Classification	Synthetic data (e.g., linear regression), Communities and Crime dataset	Experimental and Theoretical	Computational complexity in high-dimensional settings, sensitivity to choice of penalty parameters, potential scalability issues in large-scale networks
[106]	Robust to Byzantine/poisoning attacks	Reputation mechanism using Bayesian inference	Phishing method and reputation-based aggregation	CV	MNIST, Fashion-MNIST, CIFAR-10	Experimental	Scalability to larger datasets and more complex models
[105]	Robust to poisoning attacks, Byzantine attacks and label-flipping attacks	Robust mean-based aggregation	Use of a decentralized anomaly detection mechanism and robust mean aggregation	CV	MNIST, CIFAR-10	Experimental	High computational overhead, especially in large-scale FL environments
[199]	Robust to Byzantine failures	Enhanced median-based and trimmed-mean-based defense methods	Offloads Byzantine defense to programmable switches, implements enhanced median-based and trimmed-mean-based defense methods	CV	CIFAR-10, ResNet, GoogleNet, Transformer, VGG16, BERT	Experimental	Potential constraints due to limited memory and processing capabilities of switches, complexity in implementation and hardware dependency, scalability issues with increasing model size and network traffic
[254]	Robust to model poisoning attacks and Byzantine attacks	Two-stage aggregation	Leverages random noise for Byzantine aggregation, two-stage aggregation with filtering and gradient similarity	CV	MNIST, CIFAR-10	Experimental and Theoretical	Assumes the existence of a small auxiliary dataset for the server, potential challenges in scaling to larger networks, computational overhead due to the combination of DP and Byzantine resilience
[280]	Robust to model poisoning attacks and Byzantine attacks	Robust mean-based aggregation	Secure cosine similarity and incentive-based Byzantine-tolerant aggregation	CV	MNIST, CIFAR-10	Both Experimental and Theoretical	Potential computational overheads despite optimizations, especially in larger-scale implementations; needs further efficiency improvements for widespread adoption
[233]	Robust to poisoning attacks	Performance-based weighted average	FedTop algorithm: parameter normalization, performance ranking, and performance-based weighted average	CV	MNIST, CIFAR-10, YELP	Experimental and Theoretical	Limitations in dataset choosing, base model choosing, and number of malicious models
[150]	Robust to Byzantine attacks	Robust aggregation using correntropy	Maximum Correntropy Aggregation (MCA) method using fixed-point iteration	CV	MNIST, Fashion-MNIST, CIFAR-10	Both Experimental and Theoretical	Sensitivity to choice of kernel width; Limited robustness to highly sophisticated Byzantine attacks
[90]	Robust to local model poisoning attacks	Trusted coordinate-based aggregation	FL Trusted Coordinates (FLTC), which selects trusted coordinates and aggregates based on changes in direction and magnitude	CV	MNIST, Fashion-MNIST, CIFAR-10, CINIC-10, CIFAR-100	Experimental	The framework may not fully address other advanced attacks, and defense performance may degrade with larger datasets or more complex models
[50]	Robust to model poisoning attacks and adversarial attacks	FL under non-IID settings with robust aggregation techniques	Use of FedAvg, FedProx, and Scaffold algorithms	Intrusion Detection	CICIG2023	Experimental	Scalability concerns and handling of statistical heterogeneity in real-world IoT scenarios
[41]	Robust to poisoning attacks and Byzantine attacks	Confidence score-based aggregation	Scaling of gradient updates based on confidence scores derived from SoftMax probabilities	Medical Imaging	Retinal OCT Images (Kaggle)	Experimental	Needs further validation on more diverse datasets and real-world scenarios, potential impact of miscalculated confidence scores on model performance

Table 19. Modification of Model Aggregation-based Treatments to Attain Robustness in FL (Part 3)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[142]	Robust to Byzantine attacks and adversarial attacks	Robust aggregation methods	Uses coordinate-wise median and coordinate-wise trimmed mean aggregation to handle Byzantine machines and optimize models for each cluster	Simulated data	Simulated data	Both Experimental and Theoretical	Limited to strongly convex loss functions; performance might be affected by high-dimensional data and varying cluster sizes
[314]	Robust to Byzantine attacks	Byzantine-robust protocols	Introduced multiple robust estimators, including NO-REGRET, FILTERING, and GAN-based algorithms; use of bucketing and various robust estimation subroutines	CV	MNIST, Fashion-MNIST	Both Experimental and Theoretical	Higher computational complexity compared to simpler methods such as median or trimmed mean; further research needed for designing statistically optimal, computationally efficient protocols
[24]	Robust to poisoning attacks and label-flipping attacks	Generative model-based selective aggregation	Selective parameter aggregation driven by synthetic validation data from CVAEs	CV	MNIST	Experimental	Limitation in handling scenarios with a large number of malicious clients and potential overhead in communication and computation
[236]	Robust to Byzantine attacks, data poisoning, and backdoor attacks	Attention refinement with Wasserstein distance scoring	FedNAT: Using attention transfer for global aggregation and FedNAT loss function for local updates	CV and HAR	MNIST, Fashion-MNIST, HAR	Experimental	Requires clean datasets for evaluation; potential privacy concerns when collecting data; may struggle with non-IID data in extreme conditions
[19]	Robust to poisoning attacks (untargeted and targeted, including backdoor attacks)	Historical information-based recovery	Uses historical information to estimate client model updates and apply optimization strategies such as warm-up, periodic correction, abnormality fixing, and final tuning	CV and HCR	MNIST, Fashion-MNIST, Purchase, HAR	Experimental and Theoretical	Potential challenge in large-scale or highly dynamic FL systems
[273]	Robust to model poisoning and data poisoning attacks	Reputation-based robust aggregation	Two-step adversary detection: DBSCAN-based noise filtering and accuracy evaluation	CV	FEMNIST, Street-12	Experimental	Potential limitations in handling extremely non-IID data distributions
[256]	Robust to model poisoning attacks and client-wise outliers	Bootstrap Median-of-Means (bMOM) estimation	Enhances federated aggregation using a bootstrap sampling method combined with median-of-means to handle outliers and adversarial clients	CV	FEMNIST, FedCelebA, Synthetic dataset	Experimental	Computational complexity due to bootstrapping and median-of-means calculations, scalability concerns in large-scale settings, sensitivity to block size and number of clusters
[218]	Robust to Byzantine attacks, and data and model poisoning attacks	Improved Weiszfeld aggregation algorithm	Enhanced Weiszfeld algorithm for robust aggregation, CNNs for semantic feature extraction in image transmission	CV	MNIST, EMINIST	Experimental	High computational complexity in large-scale deployments, challenges in handling a high number of Byzantine users, sensitivity to the configuration of the algorithm's tunable parameters

Table 20. Hyperparameter Tuning during Model Aggregation-based Treatments to Attain Robustness in FL (Part 1)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[221]	Robust to faulty and noisy clients	Filtering-out the bad weight updates	Detecting and discarding bad updates provided by the clients via the game-theoretic approach	CV	CIFAR10, MNIST, and SPAMBASE	Experimental	No non-IID data consideration; Limited FL hyperparameter combinations tested; Increase the computation cost of the server as a large number of clients
[118]	Robust to malicious clients	Auto-weighted robust FL	They propose an objective function that minimizes the weighted sum of client empirical risk, incorporating regularization. Weights are assigned based on individual risk compared to the average risk of top-performing clients.	CV	CIFAR10, FEMINIST, and Shakespeare	Theoretical and Experimental	Smaller objective hinders low-weight clients, raising fairness issues in non-IID data. No malicious client handling assumes all honesty, not realistic in FL.
[108]	Robust to malicious updates	Modifying weights during model aggregation	Clients prioritize neighbors with higher rewards, reducing cooperation with those having lower rewards to maximize overall networked rewards	Networking		Experimental	Limited experiments performed in the classical FL context; No experiments on non-IID case
[131]	Robust to Byzantine adversarial attacks	Robust aggregation via spatial-temporal pattern analysis (STPA)	It involves four steps: (1) calculate pairwise cosine similarity of client updates; (2) cluster using agglomerative clustering and threshold partition to remove anomalous gradients; (3) aggregate gradients using Median; (4) for cross-device FL, update global gradient using historical momentum update direction	CV, and Text Classification	MNIST, Fashion-MNIST, CIFAR10, and Spambase	Experimental	Poor performance on non-IID settings; Simple gradient similarity measure (cosine distance), which tends to be unstable
[281]	Robust to Byzantine attacks	Credibility assessment based aggregation	It involves five steps: (1) assess data verification score on shared client data; (2) compute anomaly score using adaptive anomaly detection model; (3) combine both scores for client model credibility; (4) aggregate initial global model with weighted coefficients based on credibility scores; (5) train initial global model on shared clients' data for non-IID adaptation	CV	MNIST, and CIFAR10	Experimental	Rely on the shared client data, which violates the data privacy; Limited experiments performed in the classical FL context
[53]	Robust to gradient leakage and model poisoning attacks	Adversarial-aware gradient aggregation method	It computes confidence scores based on the test accuracy on a verification dataset; then, in light of confidence scores, it judges which clients are malicious	CV	CIFAR100	Experimental	No consideration of non-IID settings; Limited experimental datasets and settings; Limited baselines (just two comparison baselines); Rely on extra private shared test dataset on the server; Limited attacks considered
[110]	Robust to noisy labels	Contribution-based sample re-weighting	It transforms the problem of computing contribution via the Shapley value of each sample to a Federated Bilevel Optimization problem to realize the computation of training sample contributions; then, based on the contribution of each sample, it re-weights each samples' contribution on the model gradient update	CV	MNIST, CIFAR10, and FEMINIST	Theoretical and Experimental	Rely on public verification dataset on the server; No consideration of non-IID data; Additional complex computation workload on the server; Extra communication cost besides the normal model gradients
[11]	Robust to targeted poisoning attacks	A reputation-based defense against poisoning attacks	It (1) uses cosine-based similarity method to measure alignment level between one client and all other clients; (2) based on the cosine similarity, adjusts adaptive learning rate of client model and computes the reputation scores; (3) according to the reputation scores of each local model, it selects the final local models to take the global aggregation	CV	MNIST, CIFAR10, and Loan	Experimental	Only one baseline method considered; Unstable cosine similarity under different non-IID settings
[77]	Robust to model poisoning attacks	Isolated forest-based anomaly detection method	It: (1) constructs decision trees for local model parameters, and computes the probability of anomalies based on search path length on iTrees; (2) employs pre-trained detection module on auxiliary data to compute the anomaly scores from the isolated forest module; (3) trims malicious nodes in iTrees and aggregates the remaining client gradients to the new global model	CV and Text Classification	MNIST, and Shakespeare	Experimental	Not considering non-IID data; Limited model poisoning attacks tested; Limited FL hyperparameters settings; Rely on auxiliary data; increasing complexity to train the auxiliary detection model; High computation cost to implement the building of the isolated forest on large and complex local model architectures
[212]	Robust to adversarial attacks	Wasserstein distance-based uncertainty set generation for aggregation	It aims to optimize model performance while preserving privacy. It permits individual model parameters and privacy budget choices for each training client. By utilizing a Wasserstein distance-based ball to create an uncertainty set and linking the privacy budget to the ball's radius, the technique transforms the FL problem into an equivalent form.	Earning Prediction and Loan Classification	Adult and Loan from Kaggle	Theoretical and Experimental	Hard thresholding is considered for allocating privacy budget; Increases computation and communication costs; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[193]	Robust to malicious clients	Supervised anomaly detector-based aggregation	The central idea of this novel anomaly detector is to differentiate benign updates from malicious ones. By implementing the anomaly detector in FL, it becomes possible to effectively filter out harmful updates.	CV	MNIST, SVHN, and CIFAR10	Experimental	The supervised anomaly detector in the server requires separate dataset and training; Limited experiments performed in the classical FL context; No experiments on non-IID data
[309]	Robust to Byzantine adversaries	Aggregation with an edge-based 3-layer FL architecture	Proposes to use the existing robust aggregation rule, e.g., trimmed mean	CV	MNIST, Fashion-MNIST, and CIFAR10	Experimental	No significant contribution to the novelty of the aggregation rule; Limited experiments performed in the classical FL context; No experiments on non-IID data
[96]	Robust to noisy labels	Exploiting k -reliable neighbors with high data expertise or similarity	k -reliable neighbors can provide clean and useful information for training even when the target client's training data is unreliable due to significant label noise, satisfying the condition of data expertise of a client, and data similarity between two clients is satisfied	CV	CIFAR10, CIFAR100, and mini-WebVision	Experimental	Requires access to other clients' local models inside another client; No treatment on the models' aggregation; Limited experiments performed in the classical FL context; No experiments on non-IID data

Table 21. Hyperparameter Tuning during Model Aggregation-based Treatments to Attain Robustness in FL (Part 2)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[237]	Robust to model poisoning attacks	Penultimate layer representations-based aggregation	Utilizing the discrepancies in pairwise penultimate layer representations (PLRs) among model updates, a trust score estimation method is introduced. Each update's trust score reflects its alignment with the majority of trustworthy clients while assigning lower scores to those distant from the benign cluster. Aggregating model changes based on trust scores update the global model.	CV	FMNIST, CIFAR10, and KATHER	Experimental	Increases computation cost; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[20]	Robust to poisoning attacks	Group-wise robust aggregation	Clusters clients into different groups, trains a separate global model in each cluster, take a majority vote among the global models to classify a test input	CV and Human Activity Recognition	MNIST, CIFAR10, and Human Activity Recognition	Experimental	Requires multiple global models training; No leveraging of any prior knowledge on the learning task or the base FL algorithm when deriving our certified security levels; Limited experiments on non-IID data
[38]	Robust to Byzantine attack	Hamming distance-based aggregation	It uses a small root-dataset and server-model for bootstrapping trust	CV	MNIST, and CIFAR10	Theoretical and Experimental	Requires a clean dataset in the server; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[60]	Robust to Byzantine attack and backdoor attack	Reconstruction error probability distribution-based aggregation	To determine the reconstruction errors of updates, it employs a variational autoencoder model. Updates are declared malicious and discarded if their reconstruction mistakes are greater than the norm.	CV	MNIST, FEMNIST, and Vehicle	Experimental	Increases computation and communication costs; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[45]	Robust to local model poisoning attacks	A modified approach based on Reject on Negative Impact (RONI) and TRIM jointly identify a training dataset subset	To counter local model poisoning, RONI and TRIM have been adapted. Before computing the global model using Byzantine-robust aggregation, both defenses remove potentially harmful local models. The first defense targets models raising global error rates, and the second focuses on loss increases, validated separately.	CV	MNIST, Fashion-MNIST, CH-MNIST, and Breast Cancer Wisconsin (Diagnostic)	Experimental	Requires a clean dataset in the server; No study on targeted poisoning attacks; Limited performance to detect compromised local models; Limited experiments performed in the classical FL context; Limited experiments on non-IID data
[188]	Robust to label-flipping attack	Learning data and model association-based aggregation	Through evaluating the alignment between the server's parameters and received ones, malicious clients can be detected. Strong alignment signifies reliable data, while weak alignment implies otherwise. This filtering process excludes malicious parameters and permits comparable modifications.	CV	MNIST, FEMNIST, CIFAR10, and APTOS	Experimental	No withstand on more powerful targeted and untargated adversarial attacks; Limited experiments performed in the classical FL context; No experiments on non-IID data
[214]	Robust to Byzantine attackers	Historical distance detection-based aggregation	The approach assumes that (i) the Euclidean distance varies between benign and malicious clients, and (ii) the distances mostly resemble one another due to fewer malicious clients. Thus, the method picks a t-client combination to minimize their cumulative distances.	CV	MNIST	Experimental	No consideration of the detection of poisoning attacks under real non-IID data scenarios; No strategy for how the server excludes modified distances by malicious clients
[278]	Robust to outliers or stragglers	Unification of smooth and non-smooth aggregation	The proposed approach unifies smooth (geometric median and coordinate-wise median) and non-smooth aggregation through smoothing with a parameter by employing the Moreau envelope smoothed approximation	CV	MNIST, FEMNIST, and Sent140	Theoretical and Experimental	An in-depth exploration of neural network layer-specific aggregation is not explored; Limited experiments on non-IID data
[26]	Robust to tempered models	Hierarchical aggregation	Enhancing privacy protection, a Hierarchical Aggregation FL approach replaces the standard aggregation structure. Users split their parameters and send them to sub-aggregators. Sub-aggregators then forward user parameter parts to the main aggregator for final consolidation.	CV	Fashion-MNIST, and CIFAR10	Experimental	Requires a sub-aggregator; Aggregation protocol is not verifiable; Limited experiments performed in the classical FL context; No experiments on non-IID data
[18]	Robust to malicious clients	Ensemble federated aggregation	The procedure is applied to a subset of random clients to learn several global models. The global models are used to reach a consensus when predicting a testing example's label.	CV, and Human Activity Recognition	MNIST, Human Activity Recognition	Theoretical and Experimental	Requires more computation as multiple global models are trained; Limited experiments performed in the classical FL context; No experiments on non-IID data
[112]	Robust to malicious weights	Metrics learning and unsupervised clustering-based aggregation	During local training, the local data is encoded using metrics learning, and malicious data is precluded using the unsupervised clustering technique K-means	CV	Fashion-MNIST, and CIFAR10	Experimental	Additional access to the private data of the clients is required to preclude mislabeled data inside a client; Limited experiments performed in the classical FL context; No experiments on non-IID data
[241]	Robust to transmission outage and quantization errors	Joint bandwidth and quantization bit allocation across clients	To reduce the effects of quantization errors, the channel bandwidth and quantization bits are carefully divided among the clients, subject to wireless resource limits, constraints on transmission delays, and constraints on the probabilities of transmission outages	CV	MNIST	Theoretical and Experimental	No consideration of non-IID data issue in the theoretical proof; Very limited experiments performed in the classical FL context; No experiments on non-IID data
[207]	Robust to transmission channel effects, such as noise and fading	Tracking-based stochastic approximation and analog transmission policy of local models	By successfully averaging out the noise, tracking-based stochastic approximation (SA) iteration counteracts the noise's impact on the final convergence	CV	MNIST, and CIFAR10	Theoretical and Experimental	The proposed analog transmission may not be suitable in some application scenarios; Non-convexity is not considered during theoretical proof; Limited experiments performed in the classical FL context; Limited experiments on non-IID data

Table 22. Hyperparameter Tuning during Model Aggregation-based Treatments to Attain Robustness in FL (Part 3)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[216]	Robust to Byzantine attacks	Robust transmission and aggregation	Each group of clients is given a transmission time slot after being split into them at random. The parameter server uses the robust aggregation approach to combine the results from the various groups and then sends the results to the clients for another training session.	CV	MNIST	Theoretical and Experimental	No analysis on the effect of the imperfect channel knowledge on the proposed algorithm; Very limited experiments performed in classical FL context; No experiments on non-IID data
[268]	Robust to noisy labels	Maintains consistent decision boundaries by interchanging class-wise centroids	To ensure uniform decision boundaries, the server collaborates with local models, exchanging class-specific centroids. These centroids, representing local data features, are synchronized by the server in each communication round. This synchronization aids in forming consistent decision boundaries despite the noise distribution variations.	CV	CIFAR10, and Clothing1M	Experimental	Data privacy is compromised as the central features (class-wise centroids) of local data on each device need to be shared with the server; Limited experiments performed in classical FL context; No experiments on non-IID data
[64]	Robust to Byzantine attacks	Proactive alarming-based Byzantine-robust aggregation rules	The alarm mechanism assesses the global weights. In each communication round, clients employ their local weights and test datasets to verify the global weights. If a client detects poisoned weights, then the FL server initiates a detection process to exclude them.	CV	Fashion-MNIST, CIFAR10, Clothing1M	Experimental	Requires a local test dataset in each client; Limited experiments performed in classical FL context
[5]	Robust to Byzantine (poisoning) attacks	Distance-based outlier suppression aggregation rule	Calculates the divergence between distinct clients' local parameter updates and computes an outlier score for each using Copula-based outlier detection. These scores are transformed into normalized weights via a softmax function, influencing the global model update through weighted averaging.	Medical Imaging	CheXpert, and HAM10000	Experimental	Limited poisoning attacks tested; Limited experiments performed in classical FL context; No fairness among the clients is considered
[277]	Robust to Byzantine failures and malicious participants	Group-wise robust aggregation	Clusters model parameters into different groups and applies robust aggregation oracle within each cluster	CV	MNIST, CIFAR10	Experimental	Finding the number of clusters is costly; Limited poisoning attacks tested; Limited experiments performed in classical FL context
[318]	Robust to poisoning attacks, Byzantine attacks, free-riders	Trust-based model update strategy	Trustworthy model update using local and global model updates evaluation	Industrial Cyber-Physical Systems (CPS)	Real-world industrial CPS dataset (Gas Pipeline System, Water Storage System)	Experimental	Heavy reliance on clean auxiliary datasets, scalability challenges under complex attacks
[270]	Robust to Byzantine attacks and data poisoning attacks	Weighted aggregation	Use of backdoor triggers to identify and filter malicious local models, adjustment factors based on final layer parameters, weighted aggregation	CV	MNIST, CIFAR-10	Experimental	Potential reduction in model accuracy when backdoor triggers are embedded, requires fine-tuning of trigger proportion and detection thresholds
[125]	Robust to Byzantine attacks and model poisoning attacks	Contribution-wise Byzantine-robust aggregation rule	Dynamic assignment of honest and contribution scores to each client's gradient, combining them to generate a final score for weighted aggregation. Trimmed-normalized aggregation method to defend against scaling attacks.	CV	MNIST, Fashion MNIST, CIFAR-10	Experimental	Current approaches are focused on Byzantine robustness but may still have limitations under specific attack scenarios, such as when class imbalance is exploited. The method may require further exploration to enhance performance in non-IID settings with more complex datasets.
[139]	Robust to poisoning attacks	Geometric and qualitative metric-based defense	Uses cosine similarity, Euclidean distance, and distributed scoring by union clients	CV	MNIST, CIFAR-10	Experimental	Reliance on the quality of union client data may affect robustness under certain scenarios
[122]	Robust to Byzantine attacks	Agglomerative Hierarchical Clustering (AHC), 2PC (Two-party Computation)	Uses AHC to find benign updates, 2PC for privacy-preserving aggregation	CV	MNIST, Synthetic data	Experimental	Assumes benign-majority, potential high computational complexity, requires two non-collusive aggregators
[43]	Robust to Byzantine attacks, and data and model poisoning attacks	Server-side Mixture of Experts model	Uses a gating unit within the MoE to assign weights to client updates based on their reliability	CV	MNIST, CIFAR-10	Experimental	The method may have limited effectiveness when the server-side data is extremely scarce, and it might struggle with highly complex model structures
[267]	Robust to noisy data from clients	Bayesian framework for client utility estimation	FedTrans method using variational inference algorithm and auxiliary data	CV	CIFAR-10, Fashion-MNIST	Experimental	Dependency on auxiliary data may not cover all possible data noise patterns in clients
[99]	Robust to Byzantine attacks and model poisoning attacks	Clustering-based Krum, Sensitivity analysis of attacker count	A practical variant of Krum that estimates the number of attackers using clustering techniques and adapts Krum's aggregation accordingly	CV	MNIST	Experimental and Theoretical	Assumes benign majority; relies on accurate clustering; computational complexity with larger networks; sensitive to clustering errors
[83]	Robust to malicious clients	Hierarchical fusion of heterogeneous models, noise mitigation	HeteroArc fusion layer for aggregating heterogeneous models	CV	MNIST, CIFAR-10	Experimental	Potential scalability issues, computational overhead due to hierarchical processing, challenges in tuning for different levels of heterogeneity and noise

Table 23. Hyperparameter Tuning during Model Aggregation-based Treatments to Attain Robustness in FL (Part 4)

Ref.	Considered Robustness	Robustness Technique	Robustness Approach	Application Domain	Dataset	Evaluation Type	Potential Gap
[51]	Robust to Byzantine attacks and poisoning attacks	Differential privacy	DPFA, DPloss aggregation rule, DPVeri verification scheme	CV	MNIST, CIFAR-10	Both Experimental and Theoretical	Potential computational overhead due to the implementation of DP and verifiability; Scalability concerns with increasing client numbers
[129]	Robust to Byzantine attacks and untargeted model poisoning attacks	Clustering-based Byzantine-robust aggregation	Mini-FL framework using grouping of Non-IID features (Geo-feature, Time-feature, User-feature) for aggregation	CV	MNIST, Fashion-MNIST	Experimental	Potential limitations in extreme Non-IID scenarios and scalability to larger datasets
[175]	Robust to Byzantine attacks	Modified coordinate-wise screening, Centerpoint aggregation	Two Byzantine-resilient aggregation techniques: Modified coordinate-wise screening, Centerpoint aggregation based on discrete geometry	Not specified	Synthetic dataset	Experimental and Theoretical	Computational challenges in high-dimensional settings, difficulty in implementing centerpoint aggregation for large networks; Scalability concerns with the number of agents
[71]	Robust to poisoning attacks	Homomorphic encryption, Secret sharing, Bilinear signatures	Mask-NFRT uses double-masking and secret sharing for dropout tolerance	Medical Imaging	UCI-PT, UCI-CT	Experimental	High computational overhead due to encryption, complexity in verification process, challenges in large-scale IoMT applications
[155]	Robust to Byzantine attacks	Homomorphic encryption, Zero-knowledge proof	Combines distributed Paillier encryption with zero-knowledge proof to ensure privacy and robustness against Byzantine workers	CV	MNIST, Fashion-MNIST, EMINIST	Experimental	High computational overhead due to encryption, potential challenges in scalability, and complexity in implementation
[124]	Robust to model poisoning attacks	Aggregation algorithm modification	Krum aggregation algorithm for detecting abnormal gradients	Power grid load forecasting	Not explicitly mentioned	Both Experimental and Theoretical	Potential scalability issues and handling of heterogeneous data
[149]	Robust to Byzantine attacks	Cryptographic protocols with optimized aggregation rules	Includes a shifted DReLU function, Additive Secret Sharing, Oblivious Transfer, Pseudo-random Generator	CV	MNIST, CIFAR-10	Experimental	Relies on a trusted third-party arbitrator; high communication and computation costs in large-scale FL settings
[321]	Robust to model poisoning attacks	Client-side collaborative validation	FedValidate framework: Adjusts aggregation weights based on client credibility	CV	MNIST, CIFAR-10	Experimental	Assumes gradient visibility; limited to certain attack types
[208]	Robust to model poisoning attacks	Weighting of gradients using reputation score	Uses flip-score to detect malicious clients	CV and Text Classification	MNIST, CIFAR-10, FEMINIST, Shakespeare	Experimental, Theoretical	Dependency on the assumption that gradient flips are rare in benign conditions; may not cover scenarios where benign clients exhibit similar behavior
[62]	Robust to Byzantine/poisoning attacks	Fuzzy comprehensive evaluation method	FLEvaluate method with trust evaluation mechanism	CV	CIFAR-10	Experimental	Dependency on a clean server-side dataset for accurate trust evaluation may limit scalability
[114]	Robust to noisy labels	Parameter classification, Weighted aggregation	Classifies DNN parameters into critical and noncritical, applies different update rules, weighted aggregation based on learning efficiency	CV	CIFAR-10, Fashion-MNIST	Experimental	Potential dependency on accurate classification of parameters, may not perform well in cases with extreme noise
[219]	Robust to data poisoning attacks and malicious clients	Adaptive weighting based on Shapley value	ShapleyFL framework using surrogate federated Shapley value for client weighting	CV and Medical Imaging	CIFAR-10, Fashion-MNIST, Fed-ISIC2019 (healthcare)	Experimental	High computational cost for Shapley value calculation; Limited evaluation on complex and diverse datasets

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